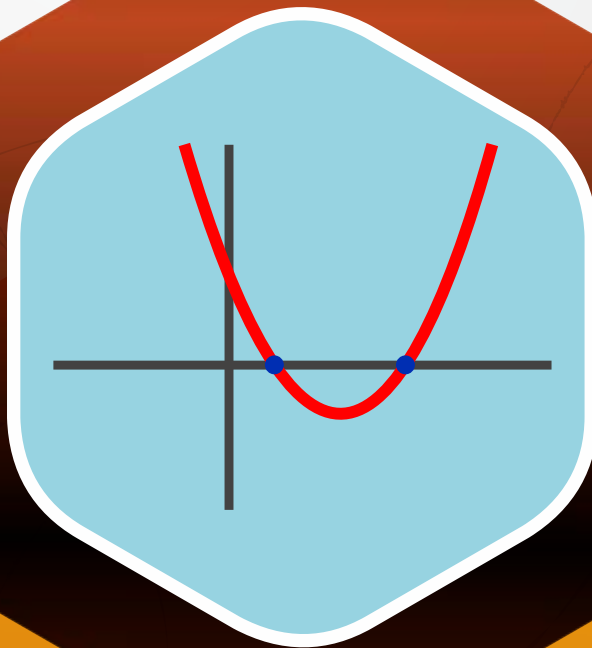


MATHEMATICS

JEE (MAIN+ADVANCED)

NURTURE COURSE



Study Material

Quadratic Equation

(English Medium)

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QUADRATIC EQUATION

1. INTRODUCTION :

The algebraic expression of the form $ax^2 + bx + c$, $a \neq 0$ is called a quadratic expression, because the highest order term in it is of second degree. Quadratic equation means, $ax^2 + bx + c = 0$. In general whenever one says zeroes of the expression $ax^2 + bx + c$, it implies roots of the equation $ax^2 + bx + c = 0$, unless specified otherwise.

A quadratic equation has exactly two roots which may be real (equal or unequal) or imaginary.

2. SOLUTION OF QUADRATIC EQUATION & RELATION BETWEEN ROOTS & CO-EFFICIENTS :

- (a) The general form of quadratic equation is $ax^2 + bx + c = 0$, $a \neq 0$.

The roots can be found in following manner :

$$a \left(x^2 + \frac{b}{a}x + \frac{c}{a} \right) = 0 \Rightarrow \left(x + \frac{b}{2a} \right)^2 + \frac{c}{a} - \frac{b^2}{4a^2} = 0$$

$$\left(x + \frac{b}{2a} \right)^2 = \frac{b^2}{4a^2} - \frac{c}{a} \Rightarrow \boxed{x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}}$$

This expression can be directly used to find the two roots of a quadratic equation.

- (b) The expression $b^2 - 4ac \equiv D$ is called the discriminant of the quadratic equation.

- (c) If α & β are the roots of the quadratic equation $ax^2 + bx + c = 0$, then :

$$(i) \alpha + \beta = -b/a \quad (ii) \alpha\beta = c/a \quad (iii) |\alpha - \beta| = \frac{\sqrt{D}}{|a|}$$

- (d) A quadratic equation whose roots are α & β is $(x - \alpha)(x - \beta) = 0$ i.e.

$$x^2 - (\alpha + \beta)x + \alpha\beta = 0 \text{ i.e. } x^2 - (\text{sum of roots})x + \text{product of roots} = 0.$$

Illustration 1 : If α, β are the roots of a quadratic equation $x^2 - 3x + 5 = 0$, then the equation whose roots are $(\alpha^2 - 3\alpha + 7)$ and $(\beta^2 - 3\beta + 7)$ is -

(A) $x^2 + 4x + 1 = 0$ (B) $x^2 - 4x + 4 = 0$ (C) $x^2 - 4x - 1 = 0$ (D) $x^2 + 2x + 3 = 0$

Solution : Since α, β are the roots of equation $x^2 - 3x + 5 = 0$

So $\alpha^2 - 3\alpha + 5 = 0$

$\beta^2 - 3\beta + 5 = 0$

$\therefore \alpha^2 - 3\alpha = -5$

$\beta^2 - 3\beta = -5$

Putting in $(\alpha^2 - 3\alpha + 7)$ & $(\beta^2 - 3\beta + 7)$ (i)

$-5 + 7, -5 + 7$

$\therefore 2$ and 2 are the roots.

$\therefore$ The required equation is $x^2 - 4x + 4 = 0$.

Ans. (B)

Illustration 2 : If α and β are the roots of $ax^2 + bx + c = 0$, find the value of $(a\alpha + b)^{-2} + (a\beta + b)^{-2}$.

Solution : We know that $\alpha + \beta = -\frac{b}{a}$ & $\alpha\beta = \frac{c}{a}$

$$\begin{aligned}(a\alpha + b)^{-2} + (a\beta + b)^{-2} &= \frac{1}{(a\alpha + b)^2} + \frac{1}{(a\beta + b)^2} \\&= \frac{a^2\beta^2 + b^2 + 2ab\beta + a^2\alpha^2 + b^2 + 2ab\alpha}{(a^2\alpha\beta + ba\beta + ba\alpha + b^2)^2} = \frac{a^2(\alpha^2 + \beta^2) + 2ab(\alpha + \beta) + 2b^2}{(a^2\alpha\beta + ab(\alpha + \beta) + b^2)^2} \\&(\alpha^2 + \beta^2 \text{ can always be written as } (\alpha + \beta)^2 - 2\alpha\beta) \\&= \frac{a^2[(\alpha + \beta)^2 - 2\alpha\beta] + 2ab(\alpha + \beta) + 2b^2}{(a^2\alpha\beta + ab(\alpha + \beta) + b^2)^2} = \frac{a^2\left[\frac{b^2 - 2ac}{a^2}\right] + 2ab\left(-\frac{b}{a}\right) + 2b^2}{\left(a^2\frac{c}{a} + ab\left(-\frac{b}{a}\right) + b^2\right)^2} = \frac{b^2 - 2ac}{a^2c^2}\end{aligned}$$

Alternatively :

If α & β are roots of $ax^2 + bx + c = 0$

then, $a\alpha^2 + b\alpha + c = 0$

$$\Rightarrow a\alpha + b = -\frac{c}{\alpha}$$

$$\text{same as } a\beta + b = -\frac{c}{\beta}$$

$$\begin{aligned}\therefore (a\alpha + b)^{-2} &= (\alpha\beta + b)^{-2} = \frac{\alpha^2}{c^2} + \frac{\beta^2}{c^2} \\&= \frac{(\alpha + \beta)^2 - 2\alpha\beta}{c^2} \\&= \frac{(-b/a)^2 - 2(c/a)}{c^2} \\&= \frac{b^2 - 2ac}{a^2c^2}\end{aligned}$$

Do yourself - 1 :

1. Find the roots of following equations :

(a) $x^2 + 3x + 2 = 0$

(b) $x^2 - 8x + 16 = 0$

(c) $x^2 - 2x - 1 = 0$

2. Find the roots of the equation $a(x^2 + 1) - (a^2 + 1)x = 0$, where $a \neq 0$.

3. Solve : $\frac{6-x}{x^2-4} = 2 + \frac{x}{x+2}$

4. If the roots of $4x^2 + 5k = (5k + 1)x$ differ by unity, then find the values of k .

5. Form a quadratic equation whose roots are the numbers $\frac{1}{10-\sqrt{72}}$ and $\frac{1}{10+6\sqrt{2}}$.
6. For what values of a is the sum of the roots of the equation $x^2 + (2 - a - a^2)x - a^2 = 0$ equal to zero?
7. For what values of a is the ratio of the roots of the equation $ax^2 - (a + 3)x + 3 = 0$ equal to 1.5?
8. The roots x_1 and x_2 of the equation $x^2 + px + 12 = 0$ are such that $x_2 - x_1 = 1$. Find p .
9. Find k in the equation $5x^2 - kx + 1 = 0$ such that the difference between the roots of the equation is unity.
10. Find p in the equation $x^2 - 4x + p = 0$ if it is known that the sum of the squares of its roots is equal to 16.
11. For what values of a is the difference between the roots of the equation $2x^2 - (a + 1)x + (a - 1) = 0$ equal to their product?
12. Express $x_1^3 + x_2^3$ in terms of the coefficients of the equation $x^2 + px + q = 0$, where x_1 and x_2 are the roots of the equation.
13. Assume that x_1 and x_2 are roots of the equation $3x^2 - ax + 2a - 1 = 0$. Calculate $x_1^3 + x_2^3$.
14. Without solving the equation $3x^2 - 5x - 2 = 0$, find the sum of the cubes of its roots.

3. NATURE OF ROOTS :

- (a) Consider the quadratic equation $ax^2 + bx + c = 0$ where $a, b, c \in \mathbb{R}$ & $a \neq 0$ then ;

$$x = \frac{-b \pm \sqrt{D}}{2a}$$

- (i) $D > 0 \Leftrightarrow$ roots are real & distinct (unequal).
- (ii) $D = 0 \Leftrightarrow$ roots are real & coincident (equal)
- (iii) $D < 0 \Leftrightarrow$ roots are non real and complex conjugate of each other.
- (iv) If $p + iq$ is one root of a quadratic equation, then the other root must be the conjugate $p - iq$ & vice versa. ($p, q \in \mathbb{R}$ & $i = \sqrt{-1}$).

- (b) Consider the quadratic equation $ax^2 + bx + c = 0$ where $a, b, c \in \mathbb{Q}$ & $a \neq 0$ then ;

- (i) If D is a perfect square of some rational number, then roots are rational.
- (ii) If $\alpha = p + \sqrt{q}$ is one root in this case, (where p is rational & $\sqrt{q}$ is an irrational) then other root will be $p - \sqrt{q}$.

Illustration 3 : If roots of the equation $(a - b)x^2 + (c - a)x + (b - c) = 0$ are equal, then

- (A) $2b = a + c$ (B) $b = \frac{2ac}{a + c}$ (C) $b^2 = ac$ (D) none of these

Solution : $(a - b)x^2 + (c - a)x + (b - c) = 0$

As roots are equal so

$$\begin{aligned} B^2 - 4AC &= 0 \Rightarrow (c - a)^2 - 4(a - b)(b - c) = 0 \Rightarrow (c - a)^2 - 4ab + 4b^2 + 4ac - 4bc = 0 \\ \Rightarrow (c - a)^2 + 4ac - 4b(c + a) + 4b^2 &= 0 \Rightarrow (c + a)^2 - 2 \cdot (2b)(c + a) + (2b)^2 = 0 \\ \Rightarrow [c + a - 2b]^2 &= 0 \Rightarrow c + a - 2b = 0 \Rightarrow c + a = 2b \end{aligned}$$

Alternative method :

$\therefore$ Sum of the coefficients = 0

Hence one root is 1 and other root is $\frac{b - c}{a - b}$.

Given that both roots are equal, so

$$1 = \frac{b - c}{a - b} \Rightarrow a - b = b - c \Rightarrow 2b = a + c$$

Illustration 4 : If equation $\frac{x^2 - bx}{ax - c} = \frac{k - 1}{k + 1}$ has roots equal in magnitude & opposite in sign, then the value of k is -

- (A) $\frac{a + b}{a - b}$ (B) $\frac{a - b}{a + b}$ (C) $\frac{a}{b} + 1$ (D) $\frac{a}{b} - 1$

Solution : Let the roots are α & $-\alpha$.

given equation is

$$(x^2 - bx)(k + 1) = (k - 1)(ax - c) \quad \{\text{Considering, } x \neq c/a \text{ \& } k \neq -1\}$$

$$\Rightarrow x^2(k + 1) - bx(k + 1) = ax(k - 1) - c(k - 1)$$

$$\Rightarrow x^2(k + 1) - bx(k + 1) - ax(k - 1) + c(k - 1) = 0$$

$$\text{Now sum of roots} = 0 \quad (\because \alpha - \alpha = 0)$$

$$\therefore b(k + 1) + a(k - 1) = 0 \Rightarrow k = \frac{a - b}{a + b}$$

Ans. (B)

Illustration 5 : If the coefficient of the quadratic equation are rational & the coefficient of x^2 is 1, then find the equation one of whose roots is $\sqrt{2} - 1$.

Solution : Irrational roots always occur in conjugational pairs.

Hence if one root is $(-1 + \sqrt{2})$, the other root will be $(-1 - \sqrt{2})$. Equation is

$$(x - (-1 + \sqrt{2}))(x - (-1 - \sqrt{2})) = 0 \Rightarrow x^2 + 2x - 1 = 0$$

Illustration 6 : Find all the integral values of a for which the quadratic equation $(x - a)(x - 10) + 1 = 0$ has integral roots.

Solution : Here the equation is $x^2 - (a + 10)x + 10a + 1 = 0$. Since integral roots will always be rational it means D should be a perfect square.

$$\text{From (i) } D = a^2 - 20a + 96.$$

$$\Rightarrow D = (a - 10)^2 - 4 \Rightarrow 4 = (a - 10)^2 - D$$

If D is a perfect square it means we want difference of two perfect square as 4 which is possible only when $(a - 10)^2 = 4$ and $D = 0$.

$$\Rightarrow (a - 10) = \pm 2 \Rightarrow a = 12, 8$$

Ans.

Illustration 7 : If D_1 and D_2 are the discriminant of two Quadratic Equations with real coefficients respectively then comment upon the nature of the roots of the Quadratic Equations under the following conditions

$$(i) D_1 + D_2 \geq 0 \quad (ii) D_1 + D_2 < 0 \quad (iii) D_1 D_2 < 0 \quad (iv) D_1 D_2 > 0 \quad (v) D_1 D_2 = 0$$

Solution : (i) If D_1 and D_2 are discriminant of two quadratic equations and $D_1 + D_2 \geq 0$, then at least one of D_1 and $D_2 \geq 0$

$\Rightarrow$ At least one of the equation has real roots.

$$(ii) D_1 + D_2 < 0 \Rightarrow \text{At least one of } D_1 \text{ and } D_2 < 0$$

$\Rightarrow$ At least one of the equation has imaginary roots.

$$(iii) \text{ If } D_1 D_2 < 0. \Rightarrow (D_1 > 0 \text{ and } D_2 < 0) \text{ or } (D_1 < 0 \text{ and } D_2 > 0).$$

Then one of the equations has real root and other equation has imaginary roots.

$$(iv) D_1 D_2 > 0$$

Case I : If $D_1 > 0$ and $D_2 > 0$, then both the equations has real roots

Case II : If $D_1 < 0$ and $D_2 < 0$, then both the equations has non real roots

$$(v) \text{ If } D_1 D_2 = 0, \text{ then } D_1 = 0 \text{ or } D_2 = 0$$

$\Rightarrow$ at least of equation has equal roots.

Illustration 8 : Let $a > 0$, $b > 0$ and $c > 0$. Then both the roots of the equation $ax^2 + bx + c = 0$ can

(a) be real and negative

(b) have negative real parts

(c) be rational numbers

(d) none of these

Solution : (abc)

$$\text{We have } D = b^2 - 4ac.$$

$$\text{If } D \geq 0, \text{ then the roots of the equation are given by } x = \frac{-b \pm \sqrt{D}}{2a}$$

As $D = b^2 - 4ac < b^2$ ($\because a > 0, c > 0$), it follows that the roots of the quadratic equation are negative.

$$\text{In case } D < 0, \text{ then the roots of the equation are given by } x = \frac{-b \pm i\sqrt{-D}}{2a}$$

which have negative real parts. ($\because$ Both a and b are positive)

Roots will be rational numbers when $\frac{b}{a}, \frac{c}{a} \in \mathbb{Q}$ and $\frac{D}{4a^2}$ is perfect square of some rational number.

4. SYMMETRIC EXPRESSIONS OF α AND β .

Let a and b be the roots of the equation $ax^2 + bx + c = 0$ such that

$f(\alpha, \beta) = f(\beta, \alpha)$ then $f(\alpha, \beta)$ denotes symmetric expression of the roots.

e.g. $f(\alpha, \beta) = \alpha^2\beta + \alpha\beta^2$

It is to be noted that every symmetric expression in α, β can be expressed in terms of two symmetric expression $\alpha + \beta$ and $\alpha\beta$

Do yourself - 2 :

- For the following equations, find the nature of the roots (real & distinct, real & coincident or imaginary).
 - $x^2 - 6x + 10 = 0$
 - $x^2 - (7 + \sqrt{3})x + 6(1 + \sqrt{3}) = 0$
 - $4x^2 + 28x + 49 = 0$
- Prove that roots of $(x - a)(x - b) = h^2$ are always real.
- For what values of a does the equation $9x^2 - 2x + a = 6 - ax$ possess equal roots ?
- Find the value of k for which the equation $(k - 1)x^2 + (k + 4)x + k + 7 = 0$ has equal roots.
- Find the values of a for which the roots of the equation $(2a - 5)x^2 - 2(a - 1)x + 3 = 0$ are equal.
- For what values of ' a ' does the quadratic equation $x^2 + (2a\sqrt{a^2 - 3})x + 4 = 0$ possess equal roots ?
- Find the least integral value of k for which the equation $x^2 - 2(k + 2)x + 12 + k^2 = 0$ has two different real roots.
- If ℓ, m are real and $\ell \neq m$, then show that the roots of $(\ell - m)x^2 - 5(\ell + m)x - 2(\ell - m) = 0$ are real and unequal.
- For what values of m does the equation $x^2 - x + m = 0$ possess no real roots ?
- For what values of c does the equation $(c - 2)x^2 + 2(c - 2)x + 2 = 0$ possess no real roots ?
- Find integral values of k for which the quadratic equation $(k - 12)x^2 + 2(k - 12)x + 2 = 0$ possess no real roots ?
- For what values of m does the equation $x^2 - x + m^2 = 0$ possess no real roots ?
- For what values of m does the equation $mx^2 - (m + 1)x + 2m - 1 = 0$ possess no real roots ?
- If $2 + \sqrt{3}$ is a root of the equation $x^2 + bx + c = 0$, where $b, c \in \mathbb{Q}$, find b, c .
- Find the value of $x^3 + x^2 - x + 22$ when $x = 1 + 2i$.
- Find the value of $x^3 - 3x^2 - 8x + 15$ when $x = 3 + i$.

17. Consider $f(x) = x^2 + bx + c$.
- (a) Find c if $x = 0$ is a root of $f(x) = 0$.
- (b) Find c if $\alpha, \frac{1}{\alpha}$ are roots of $f(x) = 0$.
- (c) Comment on sign of b & c , if $\alpha < 0 < \beta$ & $|\beta| > |\alpha|$, where α, β are roots of $f(x) = 0$.
18. Which of the following expressions in α, β will denote the symmetric functions in α, β . Give proper reasoning.
- (i) $f(\alpha, \beta) = \alpha^2 - \beta$ (ii) $f(\alpha, \beta) = \alpha^2\beta + \alpha\beta^2$
- (iii) $f(\alpha, \beta) = \alpha^3 + \beta^3$ (iv) $f(\alpha, \beta) = \frac{\alpha}{\beta} + \frac{\beta}{\alpha}$
19. If α, β are the roots of the equation $ax^2 + bx + c = 0$, find the value of
- (A) $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$ (B) $\alpha^4\beta^7 + \alpha^7\beta^4$ (C) $\left(\frac{\alpha}{\beta} - \frac{\beta}{\alpha}\right)^2$
20. Calculate $\frac{1}{x_1^3} + \frac{1}{x_2^3}$, where x_1 and x_2 are roots of the equation $2x^2 - 3ax - 2 = 0$.
21. For what value of a is the difference between the roots of the equation $(a-2)x^2 - (a-4)x - 2 = 0$ equal to 3?
22. Find the value of a for which one root of the equation $x^2 + (2a-1)x + a^2 + 2 = 0$ is twice as large as the other.
23. For what values of a is the ratio of the roots of the equation $x^2 + ax + a + 2 = 0$ equal to 2?
24. For what values of a do the roots x_1 and x_2 of the equation $x^2 - (3a+2)x + a^2 = 0$ satisfy the relation $x_1 = 9x_2$? Find the roots.
25. Find a such that one of the roots of the equation $x^2 - \frac{15}{4}x + a = 0$ is the square of the other.
26. Find all the values of a for which the sum of the roots of the equation $x^2 - 2a(x-1) - 1 = 0$ is equal to the sum of the squares of its roots.
27. Find the coefficients of the equation $x^2 + px + q = 0$ such that its roots are equal to p and q .

5. TRANSFORMATION OF THE EQUATION :

Let $ax^2 + bx + c = 0$ be a quadratic equation with two roots α and β . If we have to find an equation whose roots are $f(\alpha)$ and $f(\beta)$, i.e. some expression in α & β , then this equation can be found by finding α in terms of y . Now as α satisfies given equation, put this α in terms of y directly in the equation.

$$y = f(\alpha)$$

By transformation, $\alpha = g(y)$

$$a(g(y))^2 + b(g(y)) + c = 0$$

This is the required equation in y .

Illustration 9 : If the roots of $ax^2 + bx + c = 0$ are α and β , then find the equation whose roots are :

(a) $\frac{-2}{\alpha}, \frac{-2}{\beta}$ (b) $\frac{\alpha}{\alpha+1}, \frac{\beta}{\beta+1}$ (c) α^2, β^2

Solution : (a) $\frac{-2}{\alpha}, \frac{-2}{\beta}$

put, $y = \frac{-2}{\alpha} \Rightarrow \alpha = \frac{-2}{y}$

$a\left(\frac{-2}{y}\right)^2 + b\left(\frac{-2}{y}\right) + c = 0 \Rightarrow cy^2 - 2by + 4a = 0$

Required equation is $cx^2 - 2bx + 4a = 0$

(b) $\frac{\alpha}{\alpha+1}, \frac{\beta}{\beta+1}$

put, $y = \frac{\alpha}{\alpha+1} \Rightarrow \alpha = \frac{y}{1-y}$

$\Rightarrow a\left(\frac{y}{1-y}\right)^2 + b\left(\frac{y}{1-y}\right) + c = 0 \Rightarrow (a+c-b)y^2 + (-2c+b)y + c = 0$

Required equation is $(a+c-b)x^2 + (b-2c)x + c = 0$

(c) α^2, β^2

put $y = \alpha^2 \Rightarrow \alpha = \sqrt{y}$

$ay + b\sqrt{y} + c = 0$

$b^2y = a^2y^2 + c^2 + 2acy$

$\Rightarrow a^2y^2 + (2ac - b^2)y + c^2 = 0$

Required equation is $a^2x^2 + (2ac - b^2)x + c^2 = 0$

Illustration 10 : If the roots of $ax^3 + bx^2 + cx + d = 0$ are α, β, γ then find equation whose roots are

$\frac{1}{\alpha\beta}, \frac{1}{\beta\gamma}, \frac{1}{\gamma\alpha}$.

Solution : Put $y = \frac{1}{\alpha\beta} = \frac{\gamma}{\alpha\beta\gamma} = -\frac{a\gamma}{d}$ ($\because \alpha\beta\gamma = -\frac{d}{a}$)

Put $x = -\frac{dy}{a}$

$\Rightarrow a\left(-\frac{dy}{a}\right)^3 + b\left(-\frac{dy}{a}\right)^2 + c\left(-\frac{dy}{a}\right) + d = 0$

Required equation is $d^2x^3 - bdx^2 + acx - a^2 = 0$

Do yourself - 3 :

- If α, β are the roots of $ax^2 + bx + c = 0$, then find the equation whose roots are
 (a) $\frac{1}{\alpha^2}, \frac{1}{\beta^2}$ (b) $\frac{1}{a\alpha + b}, \frac{1}{a\beta + b}$ (c) $\alpha + \frac{1}{\beta}, \beta + \frac{1}{\alpha}$
- If α and β are the roots of $x^2 + px + q = 0$, form the equation whose roots are $(\alpha - \beta)^2$ and $(\alpha + \beta)^2$.
- If α, β are roots of $x^2 - px + q = 0$, then find the quadratic equation whose root are $(\alpha^2 - \beta^2)(\alpha^3 - \beta^3)$ and $\alpha^2\beta^3 + \alpha^3\beta^2$.

6. EQUATION VS IDENTITY :

An equation which is true for every value of the variable within the domain is called an identity, for example : $5(a - 3) = 5a - 15$, $(a + b)^2 = a^2 + b^2 + 2ab$ for all $a, b \in \mathbb{R}$.

Note : An equation of degree ≤ 2 cannot have three or more roots & if it has, it becomes an identity.

If $ax^2 + bx + c = 0$ is an identity $\Leftrightarrow a = b = c = 0$

Illustration 11 : If the equation $(\lambda^2 - 5\lambda + 6)x^2 + (\lambda^2 - 3\lambda + 2)x + (\lambda^2 - 4) = 0$ has more than two roots, then find the value of λ ?

Solution : As the equation has more than two roots so it becomes an identity. Hence

$$\lambda^2 - 5\lambda + 6 = 0 \Rightarrow \lambda = 2, 3$$

$$\text{and } \lambda^2 - 3\lambda + 2 = 0 \Rightarrow \lambda = 1, 2$$

$$\text{and } \lambda^2 - 4 = 0 \Rightarrow \lambda = 2, -2$$

$$\text{So } \lambda = 2$$

Ans. $\lambda = 2$

7. COMMON ROOTS OF TWO QUADRATIC EQUATIONS :

(a) Atleast one common root.

Let α be the common root of $ax^2 + bx + c = 0$ & $a'x^2 + b'x + c' = 0$ then

$$a\alpha^2 + b\alpha + c = 0 \text{ \& } a'\alpha^2 + b'\alpha + c' = 0. \text{ By Cramer's Rule } \frac{\alpha^2}{bc' - b'c} = \frac{\alpha}{a'c - ac'} = \frac{1}{ab' - a'b}$$

$$\text{Therefore, } \alpha = \frac{ca' - c'a}{ab' - a'b} = \frac{bc' - b'c}{a'c - ac'}$$

So the condition for a common root is $(ca' - c'a)^2 = (ab' - a'b)(bc' - b'c)$.

(b) If both roots are same then $\frac{a}{a'} = \frac{b}{b'} = \frac{c}{c'}$.

Illustration 12 : Find p and q such that $px^2 + 5x + 2 = 0$ and $3x^2 + 10x + q = 0$ have both roots in common.

Solution : $a_1 = p, b_1 = 5, c_1 = 2$ and $a_2 = 3, b_2 = 10, c_2 = q$

We know that :

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \Rightarrow \frac{p}{3} = \frac{5}{10} = \frac{2}{q} \Rightarrow p = \frac{3}{2}; q = 4$$

Illustration 13 : Find the possible value(s) of a for which the equations $x^2 + ax + 1 = 0$ and $x^2 + x + a = 0$ have atleast one common root.

Solution : Let α is a common root

$$\text{then } \alpha^2 + a\alpha + 1 = 0$$

$$\& \quad \alpha^2 + \alpha + a = 0$$

by cramer's rule

$$\frac{\alpha^2}{a^2 - 1} = \frac{\alpha}{1 - a} = \frac{1}{1 - a}$$

$$\Rightarrow (1 - a)^2 = (a^2 - 1)(1 - a)$$

$$\Rightarrow a = 1, -2$$

Illustration 14 : If the equations $x^2 - ax + b = 0$ and $x^2 - cx + d = 0$ have one root in common and second equation has equal roots, prove that $ac = 2(b + d)$

Solution : The equation $x^2 - cx + d = 0$ has equal roots.

$$\Rightarrow D = 0 \Rightarrow D = c^2 - 4d = 0 \quad \dots(i)$$

Also, the equal roots are $x = -\frac{b}{2a}$ (for $ax^2 + bx + c = 0$ having equal roots)

$$\Rightarrow x = \frac{c}{2} \text{ is the equal root of this equation.}$$

Now this should be the common root.

$$\therefore x = \frac{c}{2} \text{ will satisfy the first equation}$$

$$\Rightarrow \frac{c^2}{4} - a\left(\frac{c}{2}\right) + b = 0 \Rightarrow c^2 + 4b = 2ac$$

$$\Rightarrow 4d + 4b = 2ac$$

[Using (i)]

$$\Rightarrow 2(d + b) = ac$$

$$\text{Hence } ac = 2(b + d)$$

Illustration 15 : If $ax^2 + bx + c = 0$ and $bx^2 + cx + a = 0$ have a root in common, find the relation between a , b and c .

Solution : Solve the two equations :

$$ax^2 + bx + c = 0 \text{ and } bx^2 + cx + a = 0$$

$$\frac{x^2}{ba - c^2} = \frac{-x}{a^2 - bc} = \frac{1}{ac - b^2}$$

$$\Rightarrow (a^2 - bc)^2 = (ba - c^2)(ac - b^2)$$

$$\text{Simplify to get : } a(a^3 + b^3 + c^3 - 3abc) = 0$$

$$\Rightarrow a = 0 \text{ or } a^3 + b^3 + c^3 = 3abc$$

This is the relation between a , b and c .

Illustration 16 : Find the conditions on a, b, c and d such that equations

$$2ax^3 + bx^2 + cx + d = 0 \text{ and } 2ax^2 + 3bx + 4c = 0 \text{ have a common root.}$$

Solution :

Let ' α ' be a common root of the given equations,

$$\text{then } 2a\alpha^3 + b\alpha^2 + c\alpha + d = 0 \quad \dots(1)$$

$$\text{and } 2a\alpha^2 + 3b\alpha + 4c = 0 \quad \dots(2)$$

Multiply (2) by α and then subtract (1) from it, to get

$$2b\alpha^2 + 3c\alpha - d = 0 \quad \dots(3)$$

Now (2) and (3) are quadratic having a common root α , so

$$\frac{\alpha^2}{3bd - 12c^2} = \frac{\alpha}{8bc + 2ad} = \frac{1}{6ac - 6b^2}, \quad \alpha^2 = \frac{bd + 4c^2}{2b^2 - 2ac}, \quad \alpha = \frac{4bc + ad}{3ac - 3b^2}$$

Eliminating α from these two equations, we get

$$(4bc + ad)^2 = \frac{9}{2} (bd + 4c^2)(b^2 - ac), \text{ which is the required condition.}$$

Illustration 17 : The values of 'a' for which the equations $x^3 - 6x^2 + (6 + a)x - 6 = 0$ and $x^2 - ax + 4 = 0$ have a common root.

Solution :

Let ' α ' is the common root.

$$\alpha^3 - 6\alpha^2 + (6 + a)\alpha - 6 = 0$$

$$\alpha^2 - a\alpha + 4 = 0$$

Adding

$$\alpha^3 - 5\alpha^2 + 6\alpha - 2 = 0$$

$\Rightarrow \alpha = 1$ is one solution of above equation.

$$\begin{array}{r} \alpha^2 - 4\alpha + 2 \\ \alpha^3 - 5\alpha^2 + 6\alpha - 2 \\ \hline \alpha^3 - \alpha^2 \\ \hline -4\alpha^2 + 6\alpha - 2 \\ \alpha - 1 \quad -4\alpha^2 + 6\alpha \\ \hline \quad \quad 2\alpha - 2 \\ \quad \quad 2\alpha - 2 \\ \hline \quad \quad \quad 0 \end{array}$$

$$\Rightarrow (\alpha - 1)(\alpha^2 - 4\alpha + 2) = 0$$

$$\Rightarrow (\alpha - 1)\{\alpha - (2 - \sqrt{2})\}\{\alpha - (2 + \sqrt{2})\} = 0$$

$$\alpha = 1, \alpha = 2 - \sqrt{2} \text{ or } \alpha = 2 + \sqrt{2}$$

This common root can be 1, $2 - \sqrt{2}$, $2 + \sqrt{2}$.

We can obtain values of 'a'.

$$\text{For } \alpha = 1, 1 - a + 4 = 0 \Rightarrow a = 5$$

$$\text{For } \alpha = 2 - \sqrt{2}, (6 - 4\sqrt{2}) - a(2 - \sqrt{2}) + 4 = 0$$

$$\Rightarrow a(2 - \sqrt{2}) = 10 - 4\sqrt{2}$$

$$\Rightarrow a = \frac{10 - 4\sqrt{2}}{2 - \sqrt{2}} = \frac{(10 - 4\sqrt{2})(2 + \sqrt{2})}{(4 - 2)}$$

$$\Rightarrow a = (5 - 2\sqrt{2})(2 + \sqrt{2})$$

$$\Rightarrow a = 10 + 5\sqrt{2} - 4\sqrt{2} - 4$$

$$\Rightarrow a = 6 + \sqrt{2}$$

$$\text{For } \alpha = 2 + \sqrt{2} \Rightarrow (6 + 4\sqrt{2}) - a(2 + \sqrt{2}) + 4 = 0$$

$$\Rightarrow a = \frac{10 + 4\sqrt{2}}{2 + \sqrt{2}} = \frac{(10 + 4\sqrt{2})(2 - \sqrt{2})}{2}$$

$$\Rightarrow a = (5 + 2\sqrt{2})(2 - \sqrt{2})$$

$$\Rightarrow a = 10 - 5\sqrt{2} + 4\sqrt{2} - 4$$

$$\Rightarrow a = 6 - \sqrt{2}$$

Thus possible values of 'a' are $5, 6 \pm \sqrt{2}$.

Illustration 18 : If each pair of the following three equations $x^2 + ax + b = 0$, $x^2 + cx + d = 0$, $x^2 + ex + f = 0$ has exactly one root in common and no roots common to all three equations, then show that $(a + c + e)^2 = 4(ac + ce + ea - b - d - f)$.

Solution : Given equations are

$$x^2 + ax + b = 0 \quad \dots(1)$$

$$x^2 + cx + d = 0 \quad \dots(2)$$

$$x^2 + ex + f = 0 \quad \dots(3)$$

Let α, β be the roots of (1), β, γ be the roots of (2) and γ, α be the roots of (3).

$$\therefore \alpha + \beta = -a, \alpha\beta = b \quad \dots(4)$$

$$\beta + \gamma = -c, \beta\gamma = d \quad \dots(5)$$

$$\gamma + \alpha = -e, \gamma\alpha = f \quad \dots(6)$$

$$\begin{aligned} \text{LHS} &= (a + c + e)^2 \\ &= (-\alpha - \beta - \beta - \gamma - \gamma - \alpha)^2 \quad \{\text{from (4), (5), (6)}\} \\ &= 4(\alpha + \beta + \gamma)^2 \quad \dots(7) \end{aligned}$$

$$\begin{aligned} \text{RHS} &= 4(ac + ce + ea - b - d - f) \\ &= 4\{(\alpha + \beta)(\beta + \gamma) + (\beta + \gamma)(\gamma + \alpha) + (\alpha + \beta)(\gamma + \alpha) - \alpha\beta - \beta\gamma - \gamma\alpha\} \quad \{\text{from (4), (5), (6)}\} \\ &= 4(\alpha^2 + \beta^2 + \gamma^2 + 2\alpha\beta + 2\beta\gamma + 2\gamma\alpha) \\ &= 4(\alpha + \beta + \gamma)^2 \quad \dots(8) \end{aligned}$$

From (7) and (8),

$$(a + c + e)^2 = 4(ac + ce + ea - b - d - f)$$

Illustrations 19 : If α, β are the roots of $x^2 + px + q = 0$, and γ, δ are the roots of $x^2 + rx + s = 0$, evaluate $(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)$ in terms of p, q, r and s . Deduce the condition that the equations have a common root.

Solution :

α, β are the roots of $x^2 + px + q = 0$

$$\therefore \alpha + \beta = -p, \alpha\beta = q \quad \dots\dots(1)$$

and γ, δ are the roots of $x^2 + rx + s = 0$

$$\therefore \gamma + \delta = -r, \gamma\delta = s \quad \dots\dots(2)$$

Now, $(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)$

$$\begin{aligned} &= [\alpha^2 - \alpha(\gamma + \delta) + \gamma\delta] [\beta^2 - \beta(\gamma + \delta) + \gamma\delta] \\ &= (\alpha^2 + r\alpha + s) (\beta^2 + r\beta + s) \\ &= \alpha^2\beta^2 + r\alpha\beta(\alpha + \beta) + r^2\alpha\beta + s(\alpha^2 + \beta^2) + sr(\alpha + \beta) + s^2 \\ &= \alpha^2\beta^2 + r\alpha\beta(\alpha + \beta) + r^2\alpha\beta + s((\alpha + \beta)^2 - 2\alpha\beta) + sr(\alpha + \beta) + s^2 \\ &= q^2 - pqr + r^2q + s(p^2 - 2q) + sr(-p) + s^2 \\ &= (q - s)^2 - rpq + r^2q + sp^2 - prs \\ &= (q - s)^2 - rq(p - r) + sp(p - r) \\ &= (q - s)^2 + (p - r)(sp - rq) \end{aligned}$$

For a common root (Let $\alpha = \gamma$ or $\beta = \delta$) $\dots\dots(3)$

then $(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta) = 0 \quad \dots\dots(4)$

from (3) and (4), we get

$$(q - s)^2 + (p - r)(sp - rq) = 0$$

$\Rightarrow (q - s)^2 = (p - r)(rq - sp)$, which is the required condition.

Do yourself - 4 :

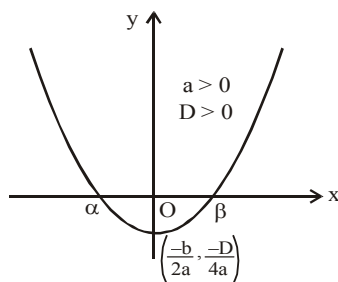
- For what values of a , the equation $(a + 3)(a - 1)x^2 + (a - 1)(4a + 3)x + (a^2 - 3a + 2) = 0$ has more than two roots.
- If $a. \frac{(x-b)(x-c)}{(a-b)(a-c)} + b. \frac{(x-c)(x-a)}{(b-c)(b-a)} + c. \frac{(x-a)(x-b)}{(c-a)(c-b)} = x$ then prove that it is an identity.
- If $x^2 + bx + c = 0$ & $2x^2 + 9x + 10 = 0$ have both roots in common, find b & c .
- If $x^2 - 7x + 10 = 0$ & $x^2 - 5x + c = 0$ have a common root, find c .
- Show that $x^2 + (a^2 - 2)x - 2a^2 = 0$ and $x^2 - 3x + 2 = 0$ have exactly one common root for all $a \in \mathbb{R}$.
- For what values of a do the equations $x^2 + ax + 1 = 0$ and $x^2 + x + a = 0$ have exactly one root in common ?
- Given two quadratic equations $x^2 - x + m = 0$ and $x^2 - x + 3m = 0, m \neq 0$. Find the value of m for which one of the roots of the second equation is equal to double the root of the first equation.

8. QUADRATIC EXPRESSION AND ITS GRAPHS :

Consider the quadratic expression, $y = ax^2 + bx + c$, $a \neq 0$ & $a, b, c \in \mathbb{R}$ then ;

- (a) The graph between x, y is always a parabola. If $a > 0$ then the shape of the parabola is concave upwards & if $a < 0$ then the shape of the parabola is concave downwards.
- (b) The graph of $y = ax^2 + bx + c$ can be divided in 6 broad categories which are as follows :
(Let the real roots of quadratic equation $ax^2 + bx + c = 0$ be α & β where $\alpha \leq \beta$).

Fig. 1

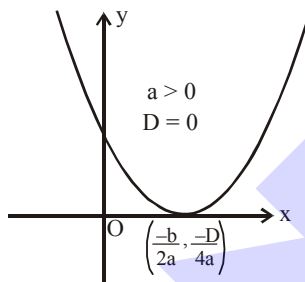


Roots are real & distinct

$$ax^2 + bx + c > 0 \quad \forall x \in (-\infty, \alpha) \cup (\beta, \infty)$$

$$ax^2 + bx + c < 0 \quad \forall x \in (\alpha, \beta)$$

Fig. 2

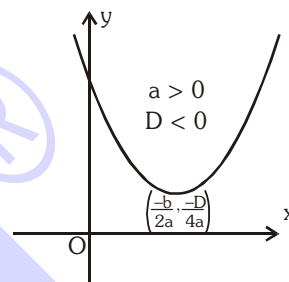


Roots are coincident

$$ax^2 + bx + c > 0 \quad \forall x \in \mathbb{R} - \{\alpha\}$$

$$ax^2 + bx + c = 0 \quad \text{for } x = \alpha = \beta$$

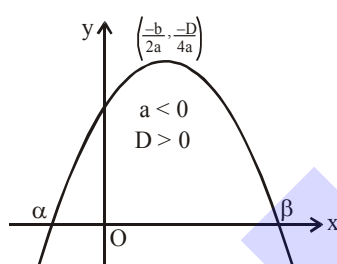
Fig. 3



Roots are complex conjugate

$$ax^2 + bx + c > 0 \quad \forall x \in \mathbb{R}$$

Fig. 4

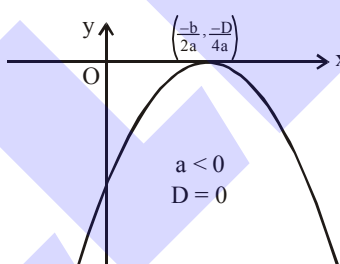


Roots are real & distinct

$$ax^2 + bx + c > 0 \quad \forall x \in (\alpha, \beta)$$

$$ax^2 + bx + c < 0 \quad \forall x \in (-\infty, \alpha) \cup (\beta, \infty)$$

Fig. 5

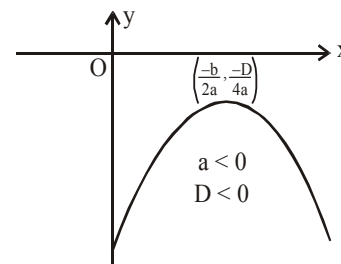


Roots are coincident

$$ax^2 + bx + c < 0 \quad \forall x \in \mathbb{R} - \{\alpha\}$$

$$ax^2 + bx + c = 0 \quad \text{for } x = \alpha = \beta$$

Fig. 6



Roots are complex conjugate

$$ax^2 + bx + c < 0 \quad \forall x \in \mathbb{R}$$

Important Note :

- (i) The quadratic expression $ax^2 + bx + c > 0$ for each $x \in \mathbb{R} \Rightarrow a > 0, D < 0$ & vice-versa (Fig. 3)
- (ii) The quadratic expression $ax^2 + bx + c < 0$ for each $x \in \mathbb{R} \Rightarrow a < 0, D < 0$ & vice-versa (Fig. 6)

9. INEQUALITIES (rational, irrational and modulus)

9.1 Rational inequalities (When numerator or denominator contains irreducible quadratic factor)

Illustration 20 : Find x such that $3x^2 - 7x + 6 < 0$

Solution : $f(x) = 3x^2 - 7x + 6$

$$D = 49 - 72 < 0 \text{ and } a > 0$$

$$\Rightarrow 3x^2 - 7x + 6 > 0 \quad \forall x \in \mathbb{R} \Rightarrow x \in \phi$$

Illustration 21: Find x such that $2x^2 + 4x + 9 > 0$

Solution : $f(x) = 2x^2 + 4x + 9$
 $a > 0$ and $D < 0$
 $\Rightarrow 2x^2 + 4x + 9 > 0 \forall x \in \mathbb{R}$
 $\Rightarrow x \in (-\infty, \infty)$

Illustration-22 : Solve for x : $\frac{(x^2 - 1)^2 + 1}{x^2 - 1} \geq 2$.

Solution : Recall :

The sum of any real number except zero and its reciprocal should not lie in $(-2, 2)$.

Case I :

When $t + \frac{1}{t} \geq 2$. This holds $\forall t > 0$ and equality holds only when $t = 1$.

Case II :

When $t + \frac{1}{t} \leq -2$. This holds $\forall t < 0$ and equality holds if $t = -1$.

Note : $x^2 - 1 \neq 0, x^2 \neq 1 \Rightarrow x \neq \pm 1$

Rewrite the given inequality $\frac{(x^2 - 1)^2 + 1}{x^2 - 1} \geq 2$

as $(x^2 - 1) + \frac{1}{(x^2 - 1)} \geq 2$

$\Rightarrow x^2 - 1 > 0 \Rightarrow (x + 1)(x - 1) > 0$



$\Rightarrow x \in (-\infty, -1) \cup (1, \infty)$

Do yourself- 5 :

Solve for $x \in \mathbb{R}$

1. $x^2 - 3x + 4 > 0$

2. $\frac{x}{x^2 + 2} < 1$

3. $\frac{x^2 - 5x + 12}{x^2 - 4x + 5} < 3$

4. $\frac{(2x - 1)(x + 3)(2 - x)(1 - x)^2}{x^4(x + 6)(x - 9)(2x^2 + 4x + 9)} < 0$

5. $\frac{7x - 17}{x^2 - 3x + 4} \geq 1$

6. $\frac{x^2 + 6x - 7}{x^2 + 1} \leq 2$

7. $\frac{2x}{x^2 - 9} \leq \frac{1}{x + 2}$

8. $\frac{x^2 - 5x + 12}{x^2 - 4x + 5} > 3$

9. $\frac{x^2 - 5x + 6}{x^2 + x + 1} < 0$

10. $\frac{x^2 + 4x + 4}{2x^2 - x - 1} > 0$

9.2. IRRATIONAL INEQUALITIES :

While solving Irrational Inequality, first we have to find largest interval of x for which left hand side and right hand side both make sense. This interval is called domain of the Inequality.

Note :

- (i) For $\sqrt{f(x)}$ to be well defined $f(x)$ must be greater than or equal to 0.
- (ii) $\sqrt{f(x)} \geq 0$ whenever defined.

Type 1 : $\sqrt{f(x)} \geq g(x)$

Case-I : $f(x) \geq 0$ and $g(x) > 0$... (1)

Above inequality become

$$f(x) \geq g^2(x) \quad \dots (2)$$

Answer in this case is intersection of solutions of (1) and (2)

Case-II : $f(x) \geq 0$ and $g(x) \leq 0$.

In this type final answer of the given Inequality is union of case I and case II.

Type 2 : $g(x) \geq \sqrt{f(x)}$

Since $\sqrt{f(x)} \geq 0$ for $\forall f(x) \geq 0$

So $g(x)$ must be non-negative

$$g(x) \geq 0 \quad \dots (1)$$

$$f(x) \geq 0 \quad \dots (2)$$

$$g^2(x) \geq f(x) \quad \dots (3)$$

Answer of the inequality is intersection of (1), (2) and (3).

Illustration-23 : Solve : $x + 1 \geq \sqrt{5 - x}$

Solution :

$$5 - x \geq 0 \text{ and } x + 1 \geq 0$$

$$\Rightarrow x \in [-1, 5] \quad \dots (1)$$

$$\text{For } x \in [-1, 5]$$

$$x + 1 \geq \sqrt{5 - x}$$

$$\Rightarrow (x + 1)^2 \geq 5 - x \Rightarrow x^2 + 3x - 4 \geq 0$$

$$\Rightarrow (x - 1)(x + 4) \geq 0 \Rightarrow x \in (-\infty, -4] \cup [1, \infty)$$

$$\Rightarrow x \in [1, 5]$$

Illustration-24 : Solve : $\sqrt{6+x} \geq 6-x$

Solution :

Case-I $6+x \geq 0$ and $6-x > 0$

$$x \in [-6, 6)$$

$$\sqrt{6+x} \geq 6-x \Rightarrow 6+x \geq 36-12x+x^2$$

$$\Rightarrow 0 \geq x^2-13x+30 \Rightarrow (x-10)(x-3) \leq 0 \Rightarrow x \in [3, 10]$$

$$x \in [3, 10] \cap [-6, 6) = [3, 6)$$

Case-II $6+x \geq 0$ and $6-x \leq 0 \Rightarrow x \in [6, \infty)$

In this case given Inequality is always true.

From case-I and case-II,

$$x \in [3, \infty).$$

Illustration 25 : Solve for x, if $\sqrt{x^2-3x+2} > x-2$

Solution :

$$\left[\begin{array}{l} \left\{ \begin{array}{l} x^2-3x+2 \geq 0 \\ x-2 \geq 0 \\ (x^2-3x+2) > (x-2)^2 \end{array} \right. \\ \text{or} \\ \left\{ \begin{array}{l} x^2-3x+2 \geq 0 \\ x-2 < 0 \end{array} \right. \end{array} \right] \Rightarrow \left[\begin{array}{l} \left\{ \begin{array}{l} (x-1)(x-2) \geq 0 \\ (x-2) \geq 0 \\ x-2 > 0 \end{array} \right. \Rightarrow x > 2 \\ \text{or} \\ \left\{ \begin{array}{l} (x-1)(x-2) \geq 0 \\ x-2 < 0 \end{array} \right. \Rightarrow x \leq 1 \end{array} \right]$$

Hence, solution set of the original inequation is $x \in \mathbb{R} - (1, 2]$

Do yourself- 6 :

Solve for x if

1. $\sqrt{x^2-x} > (x-1)$

2. $\sqrt{\frac{x-2}{1-2x}} > -1$

3. $\frac{\sqrt{x-3}}{x-2} > 0$

4. $2\sqrt{x-1} < x$

5. $\sqrt{9x-20} < x$

6. $\sqrt{5-2x} < 6x-1$

7. $\sqrt{2x-x^2} < 5-x$

8. $\sqrt{x+18} < 2-x$

9.3 Modulus inequalities

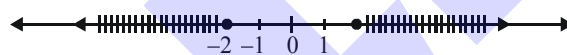
- $|x| \leq a$, if $a \geq 0$
 $\Rightarrow -a \leq x \leq a$
 if $a < 0$, then $x \in \phi$
- $|x| \geq a$
 if $a \geq 0$, then $x \in (-\infty, -a] \cup [a, \infty)$
 if $a < 0$, then $x \in (-\infty, \infty)$
- $|xy| = |x| |y|$
- $||x| - |y|| \leq |x + y| \leq |x| + |y|$
- $|x| + |y| = |x + y| \Leftrightarrow xy \geq 0$
- $|x| + |y| = |x - y| \Leftrightarrow xy \leq 0$

Illustration-26 : Represent x on the number line when

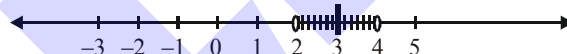
(i) $|x| \geq 2$ (ii) $|x - 3| \leq 1$ (iii) $|5 - x| < 2$

(iv) $1 \leq |x + 2| < 5$ (v) $|2x - 5| < 1$

Solution : (i) $|x| \geq 2 \Rightarrow$ distance of x from O on the number line is greater than or equal to 2.



(ii) $|x - 3| \leq 1 \Rightarrow$ distance of x from 3 is less than or equal to 1.



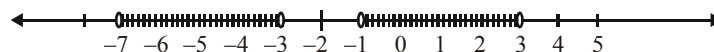
(iii) $|5 - x| < 2 \Rightarrow$ distance of x from 2 is less than 2.



(iv) $1 \leq |x + 2| < 5$

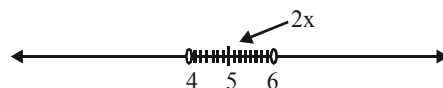
$\Rightarrow 1 \leq |x - (-2)| < 5$

$\Rightarrow$ distance of x from -2 is greater than or equal to 1 and less than 5.



(v) $|2x - 5| < 1$

$\Rightarrow$ distance of $2x$ from 5 on the number line is less than or equal to 1.



$\Rightarrow 4 < 2x < 6 \Rightarrow 2 < x < 3$

Illustration-27 : Solve following

- (i) $|x^2 - 4x| = x + 1$
 (ii) $|x^2 + 1| = |x - 2|$
 (iii) $|x^2 - 16| - 8|x - 2| = x(8 - x)$
 (iv) $a^2|x + a| + |a^2x + 1| = 1 - a^3$

where a is a real constant.

Solution :

(i) Case-1

$$x^2 - 4x \geq 0$$

$$\Rightarrow x \in (-\infty, 0] \cup [4, \infty) \quad \dots(1)$$

$$|x^2 - 4x| = x + 1$$

$$\Rightarrow x^2 - 4x = x + 1 \Rightarrow x^2 - 5x - 1 = 0$$

$$\Rightarrow x = \frac{5 \pm \sqrt{29}}{2}$$

from (1) and (2)

$$x = \frac{5 \pm \sqrt{29}}{2}$$

Case -2

where $x^2 - 4x < 0$

$$\Rightarrow x \in (0, 4) \quad \dots(3)$$

$$|x^2 - 4x| = x + 1$$

$$\Rightarrow 4x - x^2 = x + 1 \Rightarrow x^2 - 3x + 1 = 0$$

$$\Rightarrow x = \frac{3 \pm \sqrt{5}}{2} \quad \dots(4)$$

From (3) and (4)

Set of all solutions of the given equation is

$$\left\{ \frac{5 - \sqrt{29}}{2}, \frac{3 - \sqrt{5}}{2}, \frac{3 + \sqrt{5}}{2}, \frac{5 + \sqrt{29}}{2} \right\}$$

(ii) $|x^2 + 1| = |x - 2|$

As $x^2 + 1 > 0 \forall x \in \mathbb{R}$

$$\Rightarrow |x^2 + 1| = x^2 + 1, \forall x \in \mathbb{R}$$

Case-1

$$x - 2 \geq 0 \Rightarrow x \in [2, \infty)$$

$$|x^2 + 1| = |x - 2|$$

$$\Rightarrow x^2 + 1 = x - 2 \Rightarrow x^2 - x + 3 = 0$$

$$\Rightarrow x = \frac{1 \pm \sqrt{11}i}{2}$$

Case-2

$$x - 2 < 0 \Rightarrow x \in (-\infty, 2)$$

$$|x^2 + 1| = |x - 2|$$

$$\Rightarrow x^2 + 1 = -x + 2 \Rightarrow x^2 + x - 1 = 0$$

$$\Rightarrow x = \frac{-1 \pm \sqrt{5}}{2}$$

Both x belongs to $(-\infty, 2)$

$$\text{Hence set of solutions is } \left\{ \frac{-1 \pm \sqrt{5}}{2} \right\}$$

$$(iii) |x^2 - 16| - 8|x - 2| = x(8 - x)$$

Case 1 : $x \leq -4$

$$x^2 - 16 + 8x - 16 = 8x - x^2$$

$$2x^2 = 32$$

Case 2 : $x \in [-4, 2]$

$$-x^2 + 16 + 8x - 16 = 8x - x^2$$

$$x = x \text{ always true}$$

$$x \in [-4, 2]$$

Case 3 : $x \in [2, 4]$

$$-x^2 + 16 - 8x + 16 = 8x - x^2$$

$$16x = 32$$

$$x = 2$$

Case 4 : $x \geq 4$

$$x^2 - 16 - 8x + 16 = 8x - x^2$$

$$2x^2 = 16x$$

$$x = 8, 0 \text{ (rejected)}$$

$$x \in [-4, 2] \cup \{8\}$$

$$(iv) a^2|x + a| + |a^2x + 1| = 1 - a^3 \Rightarrow |a^2x + a^3| + |a^2x + 1| = 1 - a^3$$

$$\text{Let } A = a^2x + a^3, B = a^2x + 1 \text{ and } C = 1 - a^3$$

Given equation can be written as

$$|A| + |B| = C = B - A$$

$$\Rightarrow B \geq 0 \text{ and } A \leq 0$$

$$a^2x + 1 \geq 0 \text{ and } a^2x + a^3 \leq 0$$

$$\text{when } a = 0, \quad x \in \mathbb{R}$$

$$\text{when } a \neq 0, \quad x \geq \frac{-1}{a^2} \text{ and } x \leq -a \Rightarrow x \in \left[-\frac{1}{a^2}, \infty \right) \cap (-\infty, -a]$$

Case-1

$$\text{when } -\frac{1}{a^2} > -a \Rightarrow \frac{1}{a^2} < a \Rightarrow 1 < a^3 \Rightarrow a \in (1, \infty)$$

$$x \in \left[-\frac{1}{a^2}, \infty\right) \cap (-\infty, -a] = \phi$$

Case-2

$$\text{when } -a \geq -\frac{1}{a^2} \Rightarrow a \leq \frac{1}{a^2} \text{ and } a \neq 0$$

$$\Rightarrow a^3 \leq 1 \text{ and } a \neq 0 \Rightarrow a < 1 \text{ and } a \neq 0 \Rightarrow a \in (-\infty, 1] - \{0\}$$

$$x \in \left[-\frac{1}{a^2}, \infty\right) \cap (-\infty, -a]$$

$$x \in \left[-\frac{1}{a^2}, -a\right]$$

Solutions of the given equation is

$$x \in \mathbb{R} \text{ when } a = 0$$

$$x \in \phi \text{ when } a > 1$$

$$x \in \left[-\frac{1}{a^2}, -a\right] \text{ when } a \in (-\infty, 1] - \{0\}$$

Illustrations-28 : Let $A = \{x : x \in \mathbb{R}, |x| < 1\}$; $B = \{x : x \in \mathbb{R}, |x - 1| \geq 1\}$ and $A \cup B = \mathbb{R} - D$, then the set D is-

- (1) $[x : 1 < x \leq 2]$ (2) $[x : 1 \leq x < 2]$ (3) $[x : 1 \leq x \leq 2]$ (4) none of these

Solution :

$$A = \{x : x \in \mathbb{R}, -1 < x < 1\}$$

$$B = \{x : x \in \mathbb{R} : x - 1 \leq -1 \text{ or } x - 1 \geq 1\}$$

$$= \{x : x \in \mathbb{R} : x \leq 0 \text{ or } x \geq 2\}$$

$$\therefore A \cup B = \mathbb{R} - D$$

$$\text{where } D = \{x : x \in \mathbb{R}, 1 \leq x < 2\}$$

Thus (2) is the correct answer.

Illustration-29 : If $|x - 1||x - 2| = -(x^2 - 3x + 2)$, then find the interval in which x lies ?

Solution :

$$|x - 1||x - 2| = -(x - 2)(x - 1)$$

$$\Rightarrow (x - 1)(x - 2) \leq 0$$

$$\Rightarrow 1 \leq x \leq 2$$

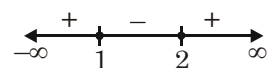


Illustration-30 : Solve : (i) $||x - 2| - 5| \geq 1$

Solution : $||x - 2| - 5| \geq 1$
 $\Rightarrow |x - 2| - 5 \geq 1$ or $|x - 2| - 5 \leq -1$
 $\Rightarrow |x - 2| \geq 6$ or $|x - 2| \leq 4$

Case-1

$$|x - 2| \geq 6$$

$$\Rightarrow x - 2 \geq 6 \text{ or } x - 2 \leq -6 \Rightarrow x \geq 8 \text{ or } x \leq -4$$

$$\Rightarrow x \in (-\infty, -4] \cup [8, \infty) \quad \dots(1)$$

Case-2

$$|x - 2| \leq 4$$

$$\Rightarrow -4 \leq x - 2 \leq 4 \Rightarrow -2 \leq x \leq 6$$

$$\Rightarrow x \in [-2, 6] \quad \dots(2)$$

From (1) and (2) $x \in (-\infty, -4] \cup [-2, 6] \cup [8, \infty)$

Illustration 31 : If x satisfies $|x - 1| + |x - 2| + |x - 3| \geq 6$, then

- (A) $0 \leq x \leq 4$ (B) $x \leq -2$ or $x \geq 4$
 (C) $x \leq 0$ or $x \geq 4$ (D) none of these

Solution :

Case I : $x \leq 1$, then
 $1 - x + 2 - x + 3 - x \geq 6 \Rightarrow x \leq 0$
 Hence $x \leq 0$...(i)

Case II : $1 < x \leq 2$, then
 $x - 1 + 2 - x + 3 - x \geq 6 \Rightarrow x \leq -2$
 But $1 < x \leq 2 \Rightarrow$ No solution. ...(ii)

Case III : $2 < x \leq 3$, then
 $x - 1 + x - 2 + 3 - x \geq 6 \Rightarrow x \geq 6$
 But $2 < x \leq 3 \Rightarrow$ No solution. ...(iii)

Case IV : $x > 3$, then
 $x - 1 + x - 2 + x - 3 \geq 6 \Rightarrow x \geq 4$
 Hence $x \geq 4$...(iv)

From (i), (ii), (iii) and (iv) the given inequality holds for $x \leq 0$ or $x \geq 4$.

Illustration 32 : Solve for x : (a) $||x - 1| + 2| \leq 4$. (b) $\frac{x-4}{x+2} \leq \left| \frac{x-2}{x-1} \right|$

Solution :

(a) $||x - 1| + 2| \leq 4 \Rightarrow -4 \leq |x - 1| + 2 \leq 4$
 $\Rightarrow -6 \leq |x - 1| \leq 2$
 $\Rightarrow |x - 1| \leq 2 \Rightarrow -2 \leq x - 1 \leq 2$
 $\Rightarrow -1 \leq x \leq 3 \Rightarrow x \in [-1, 3]$

(b) **Case 1 :** Given inequation will be satisfied for all x such that

$$\frac{x-4}{x+2} \leq 0 \Rightarrow x \in (-2, 4] - \{1\} \quad \dots\dots\dots(i)$$

(Note : $\{1\}$ is not in domain of RHS)

Case 2 : $\frac{x-4}{x+2} > 0 \Rightarrow x \in (-\infty, -2) \cup (4, \infty)$ (ii)

Given inequation becomes

$$\frac{x-2}{x-1} \geq \frac{x-4}{x+2}$$

on solving we get

$$x \in (-2, 4/5) \cup (1, \infty)$$

taking intersection with (ii) we get

$$x \in (4, \infty)$$

$$\text{or } \frac{x-2}{x-1} \leq -\frac{x-4}{x+2}$$

on solving we get

$$x \in (-2, 0] \cup (1, 5/2]$$

taking intersection with (ii) we get

$$x \in \phi$$

.....(iii)

Hence, solution of the original inequation : $x \in (-2, \infty) - \{1\}$ (taking union of (i) & (iii))

Illustration 33: The equation $|x| + \left| \frac{x}{x-1} \right| = \frac{x^2}{|x-1|}$ is always true for x belongs to

(A) $\{0\}$

(B) $(1, \infty)$

(C) $(-1, 1)$

(D) $(-\infty, \infty)$

Solution :

$$\frac{x^2}{|x-1|} = \left| x + \frac{x}{x-1} \right|$$

$$\therefore |x| + \left| \frac{x}{x-1} \right| = \left| x + \frac{x}{x-1} \right| \text{ is true only if } \left(x \cdot \frac{x}{x-1} \right) \geq 0 \Rightarrow x \in \{0\} \cup (1, \infty). \text{ Ans (A,B)}$$

Do yourself-7

Solve for $x \in \mathbb{R}$

1. $|x| \geq 10$

2. $|x-5| \leq 5$

3. $|2-x| \leq 4$

4. $1 \leq |2x-3| \leq 10$

5. $|x| < \sqrt{2}$

6. $|x| < 0$

7. $|2x| < 3$

8. $|3x| + 5 < 0$

9. $|3x-5| < 2$

10. $|1-x| \leq 2$

11. $|x| > 2$

12. $|2-7x| < 8$

13. $|2x| + 4 > 0$

14. $|2x| - 7 > 0$

10. MAXIMUM & MINIMUM VALUES OF QUADRATIC EXPRESSIONS : $y = ax^2 + bx + c$:

We know that $y = ax^2 + bx + c$ takes following form : $y = a \left[\left(x + \frac{b}{2a} \right)^2 - \frac{(b^2 - 4ac)}{4a^2} \right]$,

which is a parabola.

$$\therefore \text{ vertex } = \left(\frac{-b}{2a}, \frac{-D}{4a} \right)$$

When $a > 0$, y will take a minimum value at vertex ; $x = \frac{-b}{2a}$; $y_{\min} = \frac{-D}{4a}$

When $a < 0$, y will take a maximum value at vertex ; $x = \frac{-b}{2a}$; $y_{\max} = \frac{-D}{4a}$.

If quadratic expression $ax^2 + bx + c$ is a perfect square, then $a > 0$ and $D = 0$

Illustration 34 : The value of the expression $x^2 + 2bx + c$ will be positive for all real x if -

- (A) $b^2 - 4c > 0$ (B) $b^2 - 4c < 0$ (C) $c^2 < b$ (D) $b^2 < c$

Solution : As $a > 0$, so this expression will be positive if $D < 0$

$$\text{so } 4b^2 - 4c < 0$$

$$b^2 < c$$



Ans. (D)

Illustration 35 : The minimum value of the expression $4x^2 + 2x + 1$ is -

- (A) $1/4$ (B) $1/2$ (C) $3/4$ (D) 1

Solution : Since $a = 4 > 0$

$$\text{therefore its minimum value} = -\frac{D}{4a} = \frac{4(4)(1) - (2)^2}{4(4)} = \frac{16 - 4}{16} = \frac{12}{16} = \frac{3}{4}$$

Ans. (C)

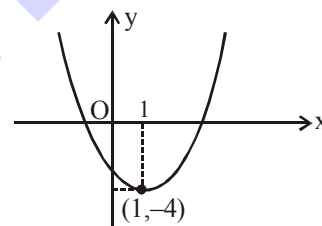
Illustration 36 : If $y = x^2 - 2x - 3$, then find the range of y when :

- (i) $x \in \mathbb{R}$ (ii) $x \in [0, 3]$ (iii) $x \in [-2, 0]$

Solution : We know that minimum value of y will occur at

$$x = -\frac{b}{2a} = -\frac{(-2)}{2 \times 1} = 1$$

$$y_{\min} = -\frac{D}{4a} = \frac{-(4 + 3 \times 4)}{4} = -4$$



- (i) $x \in \mathbb{R};$

$$y \in [-4, \infty)$$

Ans.

- (ii) $x \in [0, 3]$

$$f(0) = -3, f(1) = -4, \quad f(3) = 0$$

$$\therefore f(3) > f(0)$$

$\therefore y$ will take all the values from minimum to $f(3)$.

$$y \in [-4, 0]$$

Ans.

- (iii) $x \in [-2, 0]$

This interval does not contain the minimum value of y for $x \in \mathbb{R}$.

y will take values from $f(0)$ to $f(-2)$

$$f(0) = -3$$

$$f(-2) = 5$$

$$y \in [-3, 5]$$

Ans.

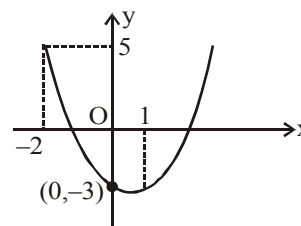


Illustration 37 : If $ax^2 + bx + 10 = 0$ does not have real & distinct roots, find the minimum value of $5a - b$.

Solution : Either $f(x) \geq 0 \forall x \in \mathbb{R}$ or $f(x) \leq 0 \forall x \in \mathbb{R}$

$$\therefore f(0) = 10 > 0 \Rightarrow f(x) \geq 0 \forall x \in \mathbb{R}$$

$$\Rightarrow f(-5) = 25a - 5b + 10 \geq 0$$

$$\Rightarrow 5a - b \geq -2$$

Ans.

Illustration-38 : Find possible values (range) of x^2 if

$$(i) x \in [1, 3] \quad (ii) x \in [-2, 0] \quad (iii) x \in [-1, 2] \quad (iv) x \in [-2, 1] \quad (v) x \in [-2, 2]$$

Solution : (i) $1 \leq x \leq 3 \Rightarrow 1 \leq x^2 \leq 9 \Rightarrow x^2 \in [1, 9]$

(ii) $-2 \leq x \leq 0 \Rightarrow 4 \geq x^2 \geq 0 \Rightarrow x^2 \in [0, 4]$

(iii) $-1 \leq x < 0$ or $0 \leq x \leq 2$

$$\Rightarrow 1 \geq x^2 > 0 \text{ or } 0 \leq x^2 \leq 4 \Rightarrow 0 \leq x^2 \leq 4 \Rightarrow x^2 \in [0, 4]$$

(iv) $-2 \leq x < 0$ or $0 \leq x \leq 1$

$$\Rightarrow 4 \geq x^2 > 0 \text{ or } 0 \leq x^2 \leq 1 \Rightarrow x^2 \in [0, 4]$$

(v) $-2 \leq x < 0$ or $0 \leq x \leq 2$

$$\Rightarrow 4 \geq x^2 > 0 \text{ or } 0 \leq x^2 \leq 4 \Rightarrow 0 \leq x^2 \leq 4 \Rightarrow x^2 \in [0, 4]$$

Illustration-39 : If $x \in \left[-\frac{3}{2}, 1\right]$ then find range of (i) $(2x - 1)^2 - 3$ (ii) $4x^3 + 1$

Solution : (i) $[-3, 13]$ (ii) $\left[-\frac{25}{2}, 5\right]$

Do yourself - 8

1. Find the minimum value of :

(a) $y = x^2 + 2x + 2$

(b) $y = 4x^2 - 16x + 15$

2. For following graphs of $y = ax^2 + bx + c$ with $a, b, c \in \mathbb{R}$, comment on the sign of :

(i) a

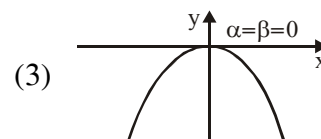
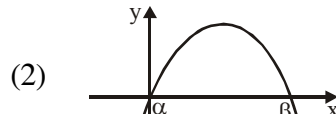
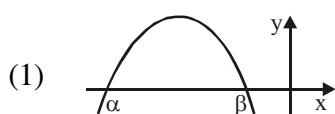
(ii) b

(iii) c

(iv) D

(v) $\alpha + \beta$

(vi) $\alpha\beta$



3. Given the roots of equation $ax^2 + bx + c = 0$ are real & distinct, where $a, b, c \in \mathbb{R}^+$, then the vertex of the graph will lie in which quadrant.

4. Find the range of 'a' for which :
 (a) $ax^2 + 3x + 4 > 0 \quad \forall x \in \mathbb{R}$ (b) $ax^2 + 4x - 2 < 0 \quad \forall x \in \mathbb{R}$
5. The trinomial $ax^2 + bx + c$ has no real roots, $a + b + c < 0$. Find the sign of the number c .
6. For what values of a do the graphs of the functions $y = 2ax + 1$ and $y = (a - 6)x^2 - 2$ not intersect?
7. For what values of k is the inequality $x^2 - (k - 3)x - k + 6 > 0$ valid for all real x ?
8. For what integral k is the inequality $x^2 - 2(4k - 1)x + 15k^2 - 2k - 7 > 0$ valid for any real x ?
9. For what values of a is the inequality $ax^2 + 2ax + 0.5 > 0$ valid for all real x ?
10. For what least integral k is the expression $(k - 2)x^2 + 8x + k + 4 > 0$ for all values of x ?
11. Find all values of a for which the inequality $(a - 3)x^2 - 2ax + 3a - 6 > 0$ is true for all $x \in \mathbb{R}$.

11. MAXIMUM & MINIMUM VALUES OF RATIONAL ALGEBRAIC EXPRESSIONS :

$$y = \frac{a_1x^2 + b_1x + c_1}{a_2x^2 + b_2x + c_2}, \frac{1}{ax^2 + bx + c}, \frac{a_1x + b_1}{a_2x^2 + b_2x + c_2}, \frac{a_1x^2 + b_1x + c_1}{a_2x + b_2} :$$

Sometime we have to find range of expression of form $\frac{a_1x^2 + b_1x + c_1}{a_2x^2 + b_2x + c_2}$.

The following procedure is used :

Step 1 : Equate the given expression to y i.e. $y = \frac{a_1x^2 + b_1x + c_1}{a_2x^2 + b_2x + c_2}$

Step 2 : By cross multiplying and simplifying, obtain a quadratic equation in x .

$$(a_1 - a_2y)x^2 + (b_1 - b_2y)x + (c_1 - c_2y) = 0$$

Step 3 : Put Discriminant ≥ 0 and solve the inequality for possible set of values of y .

Illustration 40 : If $x \in [-2, 3]$, find all possible values (Range) of

(i) $\frac{1}{x+1}$

(ii) $\frac{x+2}{x+4}$

(iii) $\frac{2x}{x-1}$

Solution :

(i) $-2 \leq x \leq 3 \Rightarrow -1 \leq x + 1 \leq 4$

(For $x + 1 = 0$, $\frac{1}{x+1}$ is undefined)

so, $-1 \leq x + 1 < 0$ or $0 < x + 1 \leq 4$

$\Rightarrow -1 \geq \frac{1}{x+1} > -\infty$ or $\infty > x+1 \geq \frac{1}{4}$

So range of $\frac{1}{x+1}$ is $(-\infty, -1] \cup \left[\frac{1}{4}, \infty\right)$

$$(ii) \quad \frac{x+2}{x+4} = 1 - \frac{2}{x+4}$$

Given $-2 \leq x \leq 3$

$$\Rightarrow 2 \leq x+4 \leq 7 \Rightarrow \frac{1}{2} \geq \frac{1}{x+4} \geq \frac{1}{7}$$

$$\Rightarrow \frac{2}{2} \geq \frac{2}{x+4} \geq \frac{2}{7} \Rightarrow -1 \leq \frac{-2}{x+4} \leq \frac{-2}{7}$$

$$\Rightarrow 0 \leq 1 - \frac{2}{x+4} \leq \frac{5}{7} \Rightarrow \text{Range} = \left[0, \frac{5}{7}\right]$$

$$(iii) \quad \frac{2x}{x-1} = 2 + \frac{2}{x-1}$$

$$\text{for } -2 \leq x \leq 3 \Rightarrow -3 \leq x-1 \leq 2$$

$$\Rightarrow -3 \leq x-1 < 0 \text{ or } 0 < x-1 \leq 2$$

$$\Rightarrow \frac{-2}{3} \geq \frac{2}{x-1} > -\infty \text{ or } \infty > \frac{2}{x-1} \geq \frac{2}{2}$$

$$\Rightarrow \frac{4}{3} \geq 2 + \frac{2}{x-1} \text{ or } 2 + \frac{2}{x-1} \geq 3$$

$$\text{so range} = \left(-\infty, \frac{4}{3}\right] \cup [3, \infty)$$

Illustration 41 : For $x \in \mathbb{R}$, find the set of values attainable by $\frac{x^2 - 3x + 4}{x^2 + 3x + 4}$.

Solution : Let $y = \frac{x^2 - 3x + 4}{x^2 + 3x + 4}$

$$x^2(y-1) + 3x(y+1) + 4(y-1) = 0$$

Case- I : $y \neq 1$

For $y \neq 1$ above equation is a quadratic equation.

So for $x \in \mathbb{R}$, $D \geq 0$

$$\Rightarrow 9(y+1)^2 - 16(y-1)^2 \geq 0 \Rightarrow 7y^2 - 50y + 7 \leq 0$$

$$\Rightarrow (7y-1)(y-7) \leq 0 \Rightarrow y \in \left[\frac{1}{7}, 7\right] - \{1\}$$

Case II : when $y = 1$

$$\Rightarrow 1 = \frac{x^2 - 3x + 4}{x^2 + 3x + 4}$$

$$\Rightarrow x^2 + 3x + 4 = x^2 - 3x + 4$$

$$\Rightarrow x = 0$$

Hence $y = 1$ for real value of x .

$$\text{so range of } y \text{ is } \left[\frac{1}{7}, 7\right]$$

Illustration 42 : Find the values of a for which the expression $\frac{ax^2 + 3x - 4}{3x - 4x^2 + a}$ assumes all real values for real values of x .

Solution :

$$\text{Let } y = \frac{ax^2 + 3x - 4}{3x - 4x^2 + a}$$

$$x^2(a + 4y) + 3(1 - y)x - (4 + ay) = 0$$

$$\text{If } x \in \mathbb{R}, D \geq 0$$

$$\Rightarrow 9(1 - y)^2 + 4(a + 4y)(4 + ay) \geq 0 \Rightarrow (9 + 16a)y^2 + (4a^2 + 46)y + (9 + 16a) \geq 0$$

$$\text{for all } y \in \mathbb{R}, (9 + 16a) > 0 \text{ \& } D \leq 0$$

$$\Rightarrow (4a^2 + 46)^2 - 4(9 + 16a)(9 + 16a) \leq 0 \Rightarrow 4(a^2 - 8a + 7)(a^2 + 8a + 16) \leq 0$$

$$\Rightarrow a^2 - 8a + 7 \leq 0 \Rightarrow 1 \leq a \leq 7$$

$$9 + 16a > 0 \text{ \& } 1 \leq a \leq 7$$

$$\text{Taking intersection, } a \in [1, 7]$$

Now, checking the boundary values of a

$$\text{For } a = 1$$

$$y = \frac{x^2 + 3x - 4}{3x - 4x^2 + 1} = -\frac{(x-1)(x+4)}{(x-1)(4x+1)}$$

$$\therefore x \neq 1 \Rightarrow y \neq -1$$

$$\Rightarrow a = 1 \text{ is not possible.}$$

$$\text{for } a = 7,$$

$$y = \frac{7x^2 + 3x - 4}{3x - 4x^2 + 7} = \frac{(7x-4)(x+1)}{(7-4x)(x+1)} \quad \therefore x \neq -1 \Rightarrow y \neq -1$$

So y will assume all real values for some real values of x .

$$\text{So } a \in (1, 7)$$

Illustration 43 : Find the values of m so that the inequality : $\left| \frac{x^2 + mx + 1}{x^2 + x + 1} \right| < 3$ holds for all $x \in \mathbb{R}$.

Solution :

We know that

$$|a| < b \Rightarrow -b < a < b \quad (\text{for } b > 0)$$

$$\text{Hence } \left| \frac{x^2 + mx + 1}{x^2 + x + 1} \right| < 3 \Rightarrow -3 < \frac{x^2 + mx + 1}{x^2 + x + 1} < 3$$

Case I :

$$\frac{x^2 + mx + 1}{x^2 + x + 1} < 3$$

$$\Rightarrow \frac{(x^2 + mx + 1) - 3(x^2 + x + 1)}{x^2 + x + 1} < 0 \Rightarrow \frac{-2x^2 + (m-3)x - 2}{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}} < 0$$

Multiplying both sides by denominator, we get :

$$\Rightarrow -2x^2 + (m-3)x - 2 < 0 \text{ (because denominator is always positive)}$$

$$\Rightarrow 2x^2 - (m-3)x + 2 > 0$$

A quadratic expression in x is always positive if coefficient of $x^2 > 0$ and $D < 0$.

$$\Rightarrow (m-3)^2 - 4(2)(2) < 0 \Rightarrow (m-3)^2 - 4^2 < 0 \Rightarrow (m-7)(m+1) < 0 \Rightarrow m \in (-1, 7) \dots(i)$$

Case II :

$$-3 < \frac{x^2 + mx + 1}{x^2 + x + 1}$$

$$\Rightarrow \frac{(x^2 + mx + 1) + 3(x^2 + x + 1)}{x^2 + x + 1} > 0 \Rightarrow 4x^2 + (m+3)x + 4 > 0$$

For this to be true for all $x \in \mathbb{R}$, coefficient of $x^2 > 0$ and $D < 0$

$$\Rightarrow (m+3)^2 - 4(4)(4) < 0$$

$$\Rightarrow (m+3-8)(m+3+8) < 0$$

$$\Rightarrow (m-5)(m+11) < 0$$

$$\Rightarrow m \in (-11, 5)$$

...(ii)

We will combine (i) and (ii) because both must be satisfied.

$$\Rightarrow \text{The common solution is } m \in (-1, 5).$$

Illustration 44 : Find the values of m for which the expression : $\frac{2x^2 - 5x + 3}{4x - m}$ can take all real values for

$$x \in \mathbb{R} - \left\{ \frac{m}{4} \right\}$$

Solution : Let $\frac{2x^2 - 5x + 3}{4x - m} = y \Rightarrow 2x^2 - (4y + 5)x + 3 + my = 0$

$$\Rightarrow \text{As } x \in \mathbb{R}, \text{ discriminant } \geq 0$$

$$\Rightarrow (4y + 5)^2 - 8(3 + my) \geq 0$$

$$\Rightarrow 16y^2 + (40 - 8m)y + 1 \geq 0 \dots(i)$$

A quadratic in y is non-negative for all values of y if coefficient of y^2 is positive and discriminant ≤ 0 .

$$\Rightarrow (40 - 8m)^2 - 4(16)(1) \leq 0 \Rightarrow (5 - m)^2 - 1 \leq 0$$

$$\Rightarrow (m - 5 - 1)(m - 5 + 1) \leq 0 \Rightarrow (m - 6)(m - 4) \leq 0$$

$$\Rightarrow m \in [4, 6] \text{ but } m = 4, 6 \text{ will be rejected as shown in Illustration 41}$$

So for the given expression to take all real values, m should take values : $m \in (4, 6)$.

Illustrations45: If $(y^2 - 5y + 3)(x^2 + x + 1) < 2x$ for all $x \in \mathbb{R}$, then find the interval in which y lies.

Solution : $(y^2 - 5y + 3)(x^2 + x + 1) < 2x, \forall x \in \mathbb{R}$

$$\Rightarrow y^2 - 5y + 3 < \frac{2x}{x^2 + x + 1}$$

$$\text{Let } \frac{2x}{x^2 + x + 1} = P$$

$$\Rightarrow px^2 + (p - 2)x + p = 0$$

$$(1) \text{ Since } x \text{ is real, } (p - 2)^2 - 4p^2 \geq 0$$

$$\Rightarrow -2 \leq p \leq \frac{2}{3}$$

$$(2) \text{ The minimum value of } 2x/(x^2 + x + 1) \text{ is } -2.$$

$$\text{So, } y^2 - 5y + 3 < -2 \Rightarrow y^2 - 5y + 5 < 0$$

$$\Rightarrow y \in \left(\frac{5 - \sqrt{5}}{2}, \frac{5 + \sqrt{5}}{2} \right)$$

Do yourself - 9 :

1. If x is real prove that the expression $\frac{8x-4}{x^2+2x-1}$ where x is real cannot have values between 2 and 4, in its range.
2. Find the range of $\frac{x^2+2x+1}{x^2+2x+7}$, where x is real
3. If x is real, then prove that $\frac{x^2-x+1}{x^2+x+1}$ lies from $\frac{1}{3}$ to 3.
4. If x be real, prove that $\frac{x}{x^2-5x+9}$ must lie between 1 and $-\frac{1}{11}$ (both inclusive).
5. If x be real, prove that $\frac{x^2+34x-71}{x^2+2x-7}$ can have no value between 5 and 9.
6. Find the greatest value of $\frac{x+2}{2x^2+3x+6}$ for real values of x .

12. LOCATION OF ROOTS :

This article deals with an elegant approach of solving problems on quadratic equations when the roots are located / specified on the number line with variety of constraints :

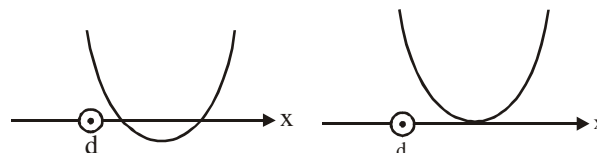
Consider the quadratic equation $ax^2 + bx + c = 0$ with $a > 0$ and let $f(x) = ax^2 + bx + c$

Type-1 :

Both roots of the quadratic equation are greater than a specific number (say d).

The necessary and sufficient condition for this are :

$$(i) D \geq 0; \quad (ii) f(d) > 0; \quad (iii) -\frac{b}{2a} > d$$



Note : When both roots of the quadratic equation are less than a specific number d than the necessary and sufficient condition will be :

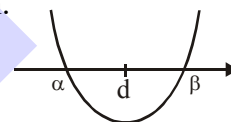
$$(i) D \geq 0; \quad (ii) f(d) > 0; \quad (iii) -\frac{b}{2a} < d$$

Type-2 :

Both roots lie on either side of a fixed number say (d). Alternatively one root is greater than 'd' and other root less than 'd' or 'd' lies between the roots of the given equation.

The necessary and sufficient condition for this are : $f(d) < 0$

Note : Consideration of discriminant is not needed.



Type-3 :

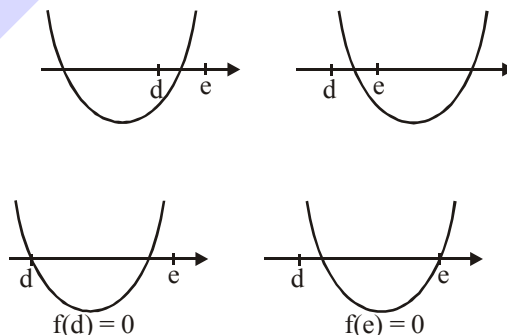
Exactly one root lies in the interval (d, e).

The necessary and sufficient condition for this are :

$$f(d) \cdot f(e) < 0$$

Note : The extremes of the intervals found by given conditions give 'd' or 'e' as the root of the equation.

Hence in this case also check for end points.



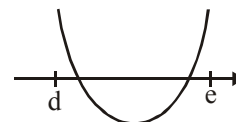
Type-4 :

When both roots are confined between the number d and e ($d < e$).

The necessary and sufficient condition for this are :

$$(i) D \geq 0; \quad (ii) f(d) > 0; \quad (iii) f(e) > 0$$

$$(iv) d < -\frac{b}{2a} < e$$



Type-5 :

One root is greater than e and the other roots is less than d ($d < e$).

The necessary and sufficient condition for this are : $f(d) < 0$ and $f(e) < 0$

Note : If $a < 0$ in the quadratic equation $ax^2 + bx + c = 0$ then we divide the whole equation by 'a'.

Now assume $x^2 + \frac{b}{a}x + \frac{c}{a}$ as $f(x)$. This makes the coefficient of x^2 positive and hence above cases are applicable.

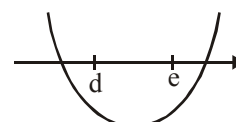


Illustration 46 : Let the quadratic equation $ax^2 + bx + c = 0$, $a, b, c \in \mathbb{R}$ has real roots then find the conditions such that

- (a) roots are equal in magnitude but opposite in sign
- (b) one root is zero other is $-b/a$
- (c) roots are reciprocal to each other
- (d) roots are of opposite signs
- (e) both roots are negative.
- (f) both roots are positive.
- (g) Greater root in magnitude is negative.
- (h) Greater root in magnitude is positive.
- (i) one root is 1 and second root is c/a or $(-b-a)/a$.

Solution

- (a) $b = 0, D > 0$
- (b) $c = 0$
- (c) $a = c, D \geq 0$
- (d) $a \cdot c < 0$
- (e) $D \geq 0, \frac{b}{a} > 0, \frac{c}{a} > 0$
- (f) $D \geq 0, \frac{b}{a} < 0, \frac{c}{a} > 0$
- (g) $\frac{b}{a} > 0, D \geq 0$
- (h) $\frac{b}{a} < 0, D \geq 0$
- (i) $a + b + c = 0$ and $a \neq 0$

Illustration 47 : Find the values of the parameter 'a' for which the roots of the quadratic equation $x^2 + 2(a-1)x + a + 5 = 0$ are

- (i) real and distinct
- (ii) equal
- (iii) opposite in sign
- (iv) equal in magnitude but opposite in sign
- (v) positive
- (vi) negative
- (vii) greater than 3
- (viii) smaller than 3
- (ix) such that both the roots lie in the interval (1, 3)

Solution :

Let $f(x) = x^2 + 2(a-1)x + a + 5 = Ax^2 + Bx + C$ (say)

$\Rightarrow A = 1, B = 2(a-1), C = a + 5.$

Also $D = B^2 - 4AC = 4(a-1)^2 - 4(a+5) = 4(a+1)(a-4)$

(i) $D > 0$

$$\Rightarrow (a+1)(a-4) > 0 \Rightarrow a \in (-\infty, -1) \cup (4, \infty).$$

(ii) $D = 0$

$$\Rightarrow (a+1)(a-4) = 0 \Rightarrow a = -1, 4.$$

(iii) This means that 0 lies between the roots of the given equation.

$$\Rightarrow f(0) < 0$$

$$\Rightarrow a + 5 < 0 \Rightarrow a < -5 \Rightarrow a \in (-\infty, -5).$$

(iv) This means that the sum of the roots is zero

$$\Rightarrow -2(a-1) = 0 \text{ and } D > 0 \text{ i.e. } a \in (-\infty, -1) \cup (4, \infty) \Rightarrow a = 1$$

which does not belong to $(-\infty, -1) \cup (4, \infty)$

$$\Rightarrow a \in \phi.$$

(v) This implies that both the roots are greater than zero

$$\Rightarrow -\frac{B}{A} > 0, \frac{C}{A} > 0, D \geq 0 \Rightarrow -(a-1) > 0, a+5 > 0, a \in (-\infty, -1] \cup [4, \infty)$$

$$\Rightarrow a < 1, -5 < a, a \in (-\infty, -1] \cup [4, \infty) \Rightarrow a \in (-5, -1].$$

(vi) This implies that both the roots are less than zero

$$\Rightarrow -\frac{B}{A} < 0, \frac{C}{A} > 0, D \geq 0 \Rightarrow -(a-1) < 0, a+5 > 0, a \in (-\infty, -1] \cup [4, \infty)$$

$$\Rightarrow a > 1, a > -5, a \in (-\infty, -1] \cup [4, \infty) \Rightarrow a \in [4, \infty).$$

(vii) In this case

$$-\frac{B}{2a} > 3, A.f(3) > 0 \text{ and } D \geq 0.$$

$$\Rightarrow -(a-1) > 3, 7a+8 > 0 \text{ and } a \in (-\infty, -1] \cup [4, \infty)$$

$$\Rightarrow a < -2, a > -8/7 \text{ and } a \in (-\infty, -1] \cup [4, \infty)$$

Since no value of 'a' can satisfy these conditions simultaneously, there can be no value of a for which both the roots will be greater than 3.

(viii) In this case

$$-\frac{B}{2a} < 3, A.f(3) > 0 \text{ and } D \geq 0.$$

$$\Rightarrow a > -2, a > -8/7 \text{ and } a \in (-\infty, -1] \cup [4, \infty) \Rightarrow a \in (-8/7, -1] \cup [4, \infty)$$

(ix) In this case

$$1 < -\frac{B}{2A} < 3, A.f(1) > 0, A.f(3) > 0, D \geq 0.$$

$$\Rightarrow 1 < -1(a-1) < 3, 3a+4 > 0, 7a+8 > 0, a \in (-\infty, -1] \cup [4, \infty)$$

$$\Rightarrow -2 < a < 0, a > -4/3, a > -8/7, a \in (-\infty, -1] \cup [4, \infty) \Rightarrow a \in \left(-\frac{8}{7}, -1\right]$$

Illustration 48 : Find value of k for which one root of equation $x^2 - (k+1)x + k^2 + k - 8 = 0$ exceeds 2 & other is less than 2.

Solution : $4 - 2(k+1) + k^2 + k - 8 < 0 \Rightarrow k^2 - k - 6 < 0$

$$(k-3)(k+2) < 0 \Rightarrow -2 < k < 3$$

Taking intersection, $k \in (-2, 3)$.

Illustration 49 : Find all possible values of a for which exactly one root of $x^2 - (a+1)x + 2a = 0$ lies in interval (0, 3).

Solution : $f(0) \cdot f(3) < 0$

$$\Rightarrow 2a(9 - 3(a+1) + 2a) < 0 \Rightarrow 2a(-a+6) < 0$$

$$\Rightarrow a(a-6) > 0 \Rightarrow a < 0 \text{ or } a > 6$$

Checking the extremes.

If $a = 0$, $x^2 - x = 0$

$$x = 0, 1$$

$$1 \in (0, 3)$$

If $a = 6$, $x^2 - 7x + 12 = 0$

$$x = 3, 4 \quad \text{But } 4 \notin (0, 3)$$

Hence solution set is

$$a \in (-\infty, 0] \cup (6, \infty)$$

Illustration 50 : Let $x^2 - (m-3)x + m = 0$ ($m \in \mathbb{R}$) be a quadratic equation. Find the value of m for which the roots of the equation are

- (i) real and distinct
- (ii) equal
- (iii) not real
- (iv) opposite in sign
- (v) equal in magnitude but opposite in sign
- (vi) positive
- (vii) negative
- (viii) such that at least one is positive
- (ix) one root is smaller than 2 and the other root is greater than 2
- (x) both the roots are greater than 2
- (xi) both the roots are smaller than 2
- (xii) exactly one root lies in the interval $(1, 2)$
- (xiii) both the roots lie in the interval $(1, 2)$
- (xiv) such that at least one root lie in the interval $(1, 2)$
- (xv) one root is greater than 2 and the other root is smaller than 1

Solution :

Let $f(x) = x^2 - (m-3)x + m$

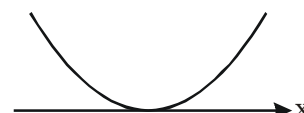
- (i) Both the roots are real and distinct $\Rightarrow D > 0$

$$\Rightarrow (m-3)^2 - 4m > 0 \Rightarrow m^2 - 10m + 9 > 0$$

$$\Rightarrow (m-1)(m-9) > 0 \Rightarrow m \in (-\infty, 1) \cup (9, \infty)$$

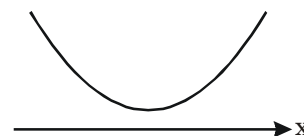


- (ii) Both the roots are equal $\Rightarrow D = 0 \Rightarrow m = 9$ or $m = 1$



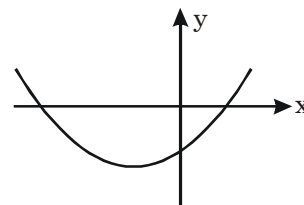
- (iii) Both the roots are imaginary $\Rightarrow D < 0$

$$(m-1)(m-9) < 0 \Rightarrow m \in (1, 9)$$



- (iv) Both the roots are opposite in sign

$$\Rightarrow f(0) < 0 \Rightarrow m < 0 \Rightarrow m \in (-\infty, 0)$$

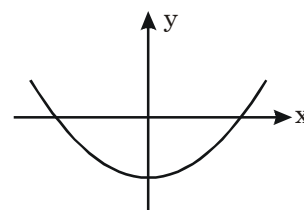


- (v) Roots are equal in magnitude but opposite in sign

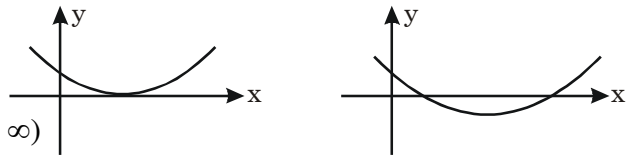
$$\Rightarrow \text{sum of roots is zero as well as } D \geq 0$$

$$\Rightarrow m \in (-\infty, 1] \cup [9, \infty) \text{ and } m-3 = 0 \text{ i.e., } m = 3$$

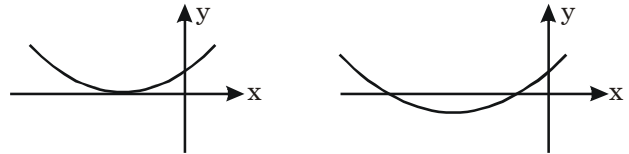
$$\Rightarrow \text{no such } m \text{ exists}$$



- (vi) Both the roots are positive $\Rightarrow D \geq 0$,
both sum and product of roots are positive
 $\Rightarrow m - 3 > 0, m > 0$ and $m \in (-\infty, 1] \cup [9, \infty)$
 $\Rightarrow m \in [9, \infty)$

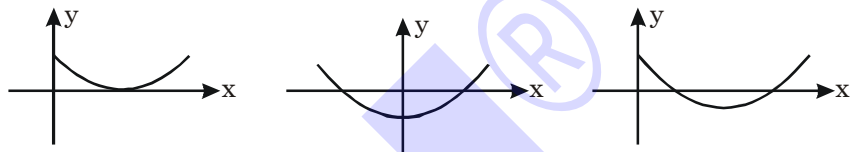


- (vii) Both the roots are negative



$\Rightarrow D \geq 0$, sum of roots is negative but product of roots is positive.
 $\Rightarrow m - 3 < 0, m > 0, m \in (-\infty, 1] \cup [9, \infty) \Rightarrow m \in (0, 1]$

- (viii) at least one root is positive

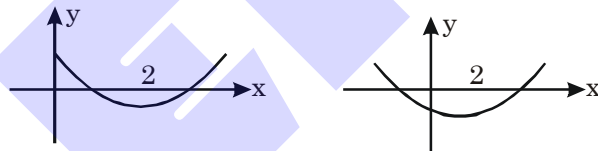


$\Rightarrow$ either one root is positive or both the roots are positive

$\Rightarrow$ union of (iv) and (vi) with $m = 0$

(i.e., one root is zero and in this case other root becomes negative) $\Rightarrow m \in (-\infty, 0) \cup [9, \infty)$

- (ix) one root is smaller than 2 and



other root is greater than 2 $\Rightarrow 2$ lies between the roots

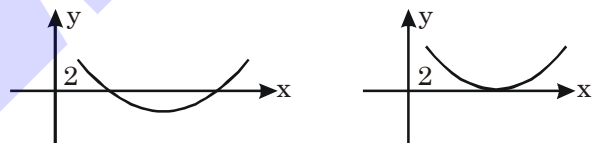
$\Rightarrow f(2) < 0 \Rightarrow 4 - 2(m - 3) + m < 0 \Rightarrow m > 10$

- (x) Both roots are greater than 2

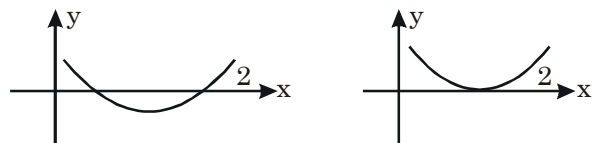
$\Rightarrow f(2) > 0, D \geq 0, \frac{-b}{2a} > 2$

$\Rightarrow m < 10$ and $m \in (-\infty, 1] \cup [9, \infty)$ and $m - 3 > 4$

$\Rightarrow m \in [9, 10)$

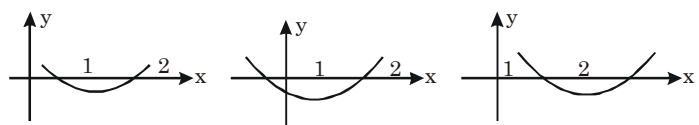


- (xi) Both the roots are smaller than 2



$\Rightarrow f(2) > 0, D \geq 0, \frac{-b}{2a} < 2 \Rightarrow m \in (-\infty, 1]$

- (xii) Exactly one root lies between (1, 2)



$\Rightarrow f(1) \cdot f(2) < 0 \Rightarrow 4(10 - m) < 0 \Rightarrow m \in (10, \infty)$

(xiii) Both roots lie in the interval (1, 2). Then

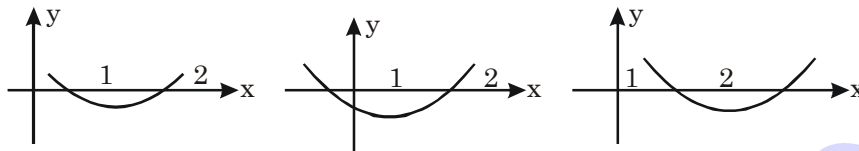
$$f(1) > 0, 1 - m + 3 + m > 0 \Rightarrow 4 > 0$$

$$f(2) > 0, 4 - 2m + 6 + m > 0 \Rightarrow m < 10$$

$$1 < \frac{m-3}{2} < 2 \Rightarrow 5 < m < 7$$

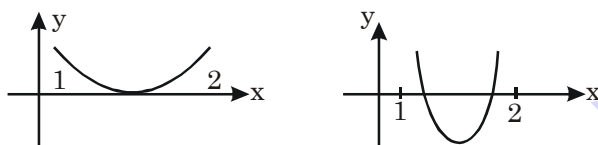
$$D \geq 0, m \in (-\infty, 1] \cup [9, \infty) \Rightarrow \text{no solution}$$

(xiv) **Case I :** Exactly one root lies in (1, 2)



$$f(1) \cdot f(2) < 0 \Rightarrow m > 10$$

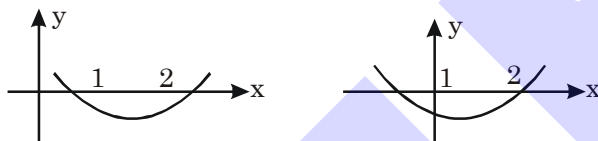
Case II : Both roots lie in (1, 2)



From (xiii) $m \in \phi$

At least one root lies in (1, 2) $\Rightarrow m \in (10, \infty)$

(xv) For one root greater than 2 and other is smaller than 1, conditions are



$$f(1) < 0 \quad \dots(1)$$

$$\text{and } f(2) < 0 \quad \dots(2)$$

From (1), $f(1) < 0$, but $f(1) = 4$ which is not possible

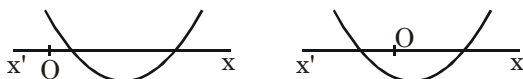
Thus no such 'm' exists.

Illustration 51 : Find the value(s) of 'a' for which $ax^2 + (a-3)x + 1 < 0$ for at least one positive x.

Solution : Let $f(x) = ax^2 + (a-3)x + 1$

Case I :

If $a > 0$, then $f(x)$ will be negative only for those values of x, which lie between the roots. From the graph we can see that, $f(x)$ will be less than zero for at least one positive real x, when $f(x) = 0$ has distinct roots and at least one of these roots is positive real root.



For this $D > 0$, i.e., $(a-3)^2 - 4a > 0$

$$\Rightarrow a < 1 \text{ or } a > 9 \quad \dots(1)$$

It is easy to see that for figure 1, $a < 3$ and figure 2 is not possible as $f(0) = 1$.

Hence for case-I $0 < a < 1$.

Case II :

If $a < 0$, then $f(x)$ will certainly be negative for infinitely many positive x . Thus $a \in (-\infty, 0)$.

Case III :

if $a = 0$, $f(x) = -3x + 1 \Rightarrow f(x) < 0 \forall x > 1/3$

Hence the required set of values of 'a' is $(-\infty, 1)$

Illustration 52 : Find the values of 'a' for which $4^t - (a - 4)2^t + \frac{9}{4}a < 0 \forall t \in (1, 2)$.

Solution : Let $2^t = x$ and $f(x) = x^2 - (a - 4)x + \frac{9}{4}a$

We want

$$f(x) < 0 \forall x \in (2^1, 2^2) \text{ i.e., } \forall x \in (2, 4)$$

Since we want $f(x) < 0 \forall x \in (2, 4)$, one of the roots of $f(x)$ should be less than or equal to 2 and the other must be greater than or equal to 4

i.e., $f(2) \leq 0$ and $f(4) \leq 0 \Rightarrow a \leq -48$ and $a \geq 128/7$, which is not possible.

Hence no such 'a' exists.

Illustration 53 : Find the value(s) of 'a' for which the inequality $\tan^2 x + (a + 1)\tan x - (a - 3) < 0$, is true for at least one $x \in \left(0, \frac{\pi}{2}\right)$.

Solution : The required condition will be satisfied if

(i) The quadratic expression (quadratic in $\tan x$)

$$f(x) = \tan^2 x + (a + 1)\tan x - (a - 3) \text{ has positive discriminant, and}$$

(ii) At least one root of $f(x) = 0$ is positive, as $\tan x > 0, \forall x \in (0, \pi/2)$

$$\text{For (i) Discriminant} > 0 \Rightarrow (a + 1)^2 + 4(a - 3) > 0$$

$$\Rightarrow a > 2\sqrt{5} - 3 \text{ or } a < -(2\sqrt{5} + 3)$$

$$\text{For (ii), we first find the condition, that both the roots of } t^2 + (a + 1)t - (a - 3) = 0$$

($t = \tan x$) are non-positive for which

$$\text{Sum of roots} < 0 \text{ product of roots} \geq 0$$

$$\Rightarrow -(a + 1) < 0 \text{ and } -(a - 3) \geq 0 \Rightarrow -1 < a \leq 3$$

$$\text{Condition (ii) will be fulfilled if } a \leq -1 \text{ or } a > 3 \quad \dots(2)$$

Required values of a is given by intersection of (1) and (2)

$$\text{Hence } a \in (-\infty, -(2\sqrt{5} + 3)) \cup (3, \infty)$$

Do yourself - 10 :

- If α, β are roots of $7x^2 + 9x - 2 = 0$, find their position with respect to following ($\alpha < \beta$) :
(a) -3 (b) 0 (c) 1
- If $a > 1$, roots of the equation $(1 - a)x^2 + 3ax - 1 = 0$ are -
(a) one positive one negative (b) both negative
(c) both positive (d) both non-real
- Find the set of value of a for which the roots of the equation $x^2 - 2ax + a^2 + a - 3 = 0$ are less than 3.
- If α, β are the roots of $x^2 - 3x + a = 0$, $a \in \mathbb{R}$ and $\alpha < 1 < \beta$, then find the values of a .
- If α, β are roots of $4x^2 - 16x + \lambda = 0$, $\lambda \in \mathbb{R}$ such that $1 < \alpha < 2$ and $2 < \beta < 3$, then find the range of λ .
- For what values of a does the equation $(2 - x)(x + 1) = a$ possess real and positive roots ?

13. GENERAL QUADRATIC EXPRESSION IN TWO VARIABLES :

$f(x, y) = ax^2 + 2hxy + by^2 + 2gx + 2fy + c$ may be resolved into two linear factors if ;

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0 \quad \text{OR} \quad \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$$

Illustration 54 : If $x^2 + 2xy + 2x + my - 3$ have two linear factor then m is equal to -

- (A) 6, 2 (B) -6, 2 (C) 6, -2 (D) -6, -2

Solution : Here $a=1, h=1, b=0, g=1, f=m/2, c=-3$

$$\text{So } \Delta = 0 \Rightarrow \begin{vmatrix} 1 & 1 & 1 \\ 1 & 0 & m/2 \\ 1 & m/2 & -3 \end{vmatrix} = 0$$

$$\Rightarrow -\frac{m^2}{4} - (-3 - m/2) + m/2 = 0 \quad \Rightarrow -\frac{m^2}{4} + m + 3 = 0$$

$$\Rightarrow m^2 - 4m - 12 = 0 \quad \Rightarrow m = -2, 6$$

Ans. (C)

Do yourself - 11 :

1. Find the value of k for which the expression $x^2 + 2xy + ky^2 + 2x + k = 0$ can be resolved into two linear factors.
2. For what values of m will the expression $y^2 + 2xy + 2x + my - 3$ be capable of resolution into two rational factors ?
3. Find the value of m which will make $2x^2 + mxy + 3y^2 - 5y - 2$ equivalent to the product of two linear factors.
4. Find the condition that the expression $\ell x^2 + mxy + ny^2$, $\ell'x^2 + m'xy + n'y^2$ may have a common linear factor.
5. Find the condition that the expression $ax^2 + 2hxy + by^2$, $a'x^2 + 2h'xy + b'y^2$ may be respectively divisible by factors of the form $y - mx$, $my + x$.
6. If x and y are two real quantities connected by the equation $9x^2 + 2xy + y^2 - 92x - 20y + 244 = 0$, then show that x will lie between 3 and 6. and y between 1 and 10.

14. THEORY OF EQUATIONS :

Let $\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n$ are roots of the equation, $f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_{n-1}x + a_n = 0$, where $a_0, a_1, \dots, a_n$ are constants and $a_0 \neq 0$.

$$f(x) = a_0(x - \alpha_1)(x - \alpha_2)(x - \alpha_3) \dots (x - \alpha_n)$$

$$\therefore a_0x^n + a_1x^{n-1} + \dots + a_{n-1}x + a_n = a_0(x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_n)$$

Comparing the coefficients of like powers of x , we get

$$\sum \alpha_i = -\frac{a_1}{a_0} = S_1 \text{ (say)}$$

$$\text{or } S_1 = -\frac{\text{coefficient of } x^{n-1}}{\text{coefficient of } x^n}$$

$$S_2 = \sum_{i \neq j} \alpha_i \alpha_j = (-1)^2 \frac{a_2}{a_0}$$

$$S_3 = \sum_{i \neq j \neq k} \alpha_i \alpha_j \alpha_k = (-1)^3 \frac{a_3}{a_0}$$

$\vdots$

$$S_n = \alpha_1 \alpha_2 \dots \alpha_n = (-1)^n \frac{a_n}{a_0} = (-1)^n \frac{\text{constant term}}{\text{coefficient of } x^n}$$

where S_k denotes the sum of the product of root taken k at a time.

Quadratic equation : If α, β are the roots of the quadratic equation $ax^2 + bx + c = 0$,

$$\text{then } \alpha + \beta = -\frac{b}{a} \text{ and } \alpha\beta = \frac{c}{a}$$

Cubic equation : If α, β, γ are roots of a cubic equation $ax^3 + bx^2 + cx + d = 0$, then

$$\alpha + \beta + \gamma = -\frac{b}{a}, \quad \alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a} \text{ and } \alpha\beta\gamma = -\frac{d}{a}$$

Note :

- (i) If α is a root of the equation $f(x) = 0$, then the polynomial $f(x)$ is exactly divisible by $(x - \alpha)$ or $(x - \alpha)$ is a factor of $f(x)$ and conversely.
- (ii) Every equation of n th degree ($n \geq 1$) has exactly n roots & if the equation has more than n roots, it is an identity.
- (iii) If the coefficients of the equation $f(x) = 0$ are all real and $\alpha + i\beta$ is its root, then $\alpha - i\beta$ is also a root. i.e. **imaginary roots occur in conjugate pairs.**
- (iv) If the coefficients in the equation are all rational & $\alpha + \sqrt{\beta}$ is one of its roots, then $\alpha - \sqrt{\beta}$ is also a root where $\alpha, \beta \in \mathbb{Q}$ & β is not a perfect square.
- (v) If there be any two real numbers 'a' & 'b' such that $f(a)$ & $f(b)$ are of opposite signs, then $f(x) = 0$ must have at least one real root between 'a' and 'b'.
- (vi) Every equation $f(x) = 0$ of degree odd and leading coefficient positive has at least one real root of a sign opposite to that of its non zero constant term.

Illustration 55: If two roots are equal, find the roots of $4x^3 + 20x^2 - 23x + 6 = 0$.

Solution :

Let roots be α, α and β

$$\therefore \alpha + \alpha + \beta = -\frac{20}{4} \Rightarrow 2\alpha + \beta = -5 \quad \dots\dots\dots (i)$$

$$\therefore \alpha \cdot \alpha + \alpha\beta + \alpha\beta = -\frac{23}{4} \Rightarrow \alpha^2 + 2\alpha\beta = -\frac{23}{4} \text{ \& } \alpha^2\beta = -\frac{6}{4}$$

from equation (i)

$$\alpha^2 + 2\alpha(-5 - 2\alpha) = -\frac{23}{4} \Rightarrow \alpha^2 - 10\alpha - 4\alpha^2 = -\frac{23}{4} \Rightarrow 12\alpha^2 + 40\alpha - 23 = 0$$

$$\therefore \alpha = 1/2, -\frac{23}{6}$$

$$\text{when } \alpha = \frac{1}{2}$$

$$\alpha^2\beta = \frac{1}{4}(-5 - 1) = -\frac{3}{2}$$

$$\text{when } \alpha = -\frac{23}{6} \Rightarrow \alpha^2\beta = \frac{23 \times 23}{36} \left(-5 - 2 \times \left(-\frac{23}{6} \right) \right) \neq -\frac{3}{2} \Rightarrow \alpha = \frac{1}{2} \quad \beta = -6$$

$$\text{Hence roots of equation} = \frac{1}{2}, \frac{1}{2}, -6 \quad \text{Ans.}$$

Illustration 56 : If α, β, γ are the roots of $x^3 - px^2 + qx - r = 0$, find :

$$(i) \quad \sum \alpha^3 \qquad (ii) \quad \alpha^2(\beta + \gamma) + \beta^2(\gamma + \alpha) + \gamma^2(\alpha + \beta)$$

Solution : We know that $\alpha + \beta + \gamma = p$

$$\alpha\beta + \beta\gamma + \gamma\alpha = q$$

$$\alpha\beta\gamma = r$$

$$(i) \quad \alpha^3 + \beta^3 + \gamma^3 = 3\alpha\beta\gamma + (\alpha + \beta + \gamma)\{(\alpha + \beta + \gamma)^2 - 3(\alpha\beta + \beta\gamma + \gamma\alpha)\} \\ = 3r + p\{p^2 - 3q\} = 3r + p^3 - 3pq$$

$$(ii) \quad \alpha^2(\beta + \gamma) + \beta^2(\gamma + \alpha) + \gamma^2(\alpha + \beta) = \alpha^2(p - \alpha) + \beta^2(p - \beta) + \gamma^2(p - \gamma) \\ = p(\alpha^2 + \beta^2 + \gamma^2) - 3r - p^3 + 3pq = p(p^2 - 2q) - 3r - p^3 + 3pq = pq - 3r$$

Illustration 57 : If $b^2 < 2ac$ and $a, b, c, d \in \mathbb{R}$, then prove that $ax^3 + bx^2 + cx + d = 0$ has exactly one real root.

Solution : Let α, β, γ be the roots of $ax^3 + bx^2 + cx + d = 0$

$$\text{Then } \alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \beta\gamma + \gamma\alpha = \frac{c}{a}$$

$$\alpha\beta\gamma = \frac{-d}{a}$$

$$\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \gamma\alpha) = \frac{b^2}{a^2} - \frac{2c}{a} = \frac{b^2 - 2ac}{a^2}$$

$\Rightarrow \alpha^2 + \beta^2 + \gamma^2 < 0$, which is not possible if all α, β, γ are real. So atleast one root is non-real, but complex roots occurs in pair. Hence given cubic equation has two non-real and one real roots.

Illustration 58 : If the sum of two roots of the equation $x^3 - px^2 + qx - r = 0$ is zero, then prove that $pq = r$.

Solution : Let the roots of the given equation by α, β, γ such that $\alpha + \beta = 0$, Then,

$$\alpha + \beta + \gamma = -\frac{(-p)}{1} \Rightarrow \alpha + \beta + \gamma = p \Rightarrow \gamma = p \quad [\because \alpha + \beta = 0]$$

But γ is a root of the given equation. Therefore,

$$\gamma^3 - p\gamma^2 + q\gamma - r = 0 \Rightarrow p^3 - p^3 + qp - r = 0 \Rightarrow pq = r$$

Illustration 59 : Find the condition that the roots of the equation $x^3 - px^2 + qx - r = 0$ may be in A.P.

Solution : Let the roots of the given equation be $a - d, a, a + d$.

Then,

$$(a - d) + a + (a + d) = \frac{-(-p)}{1} \Rightarrow a = p/3$$

Since a is a root of the given equation. Therefore,

$$a^3 - pa^2 + qa - r = 0 \Rightarrow \frac{p^3}{27} - \frac{p^3}{9} + \frac{qp}{3} - r = 0 \Rightarrow 2p^3 - 9pq + 27r = 0$$

This is the required condition.

Illustration 60 : If $x^2 + x + 1$ is a factor of $ax^3 + bx^2 + cx + d$, then find the real root of

$$ax^3 + bx^2 + cx + d = 0.$$

Solution : $x^2 + x + 1$ is a factor of $ax^3 + bx^2 + cx + d$. So roots of $x^2 + x + 1 = 0$ are also the roots of the equation $ax^3 + bx^2 + cx + d = 0$.

Let third roots of $ax^3 + bx^2 + cx + d$ is α .

Now we can write,

$$ax^3 + bx^2 + cx + d = a(x^2 + x + 1)(x - \alpha)$$

Comparing constant term on both sides, we get : $d = -a\alpha$

Hence, the real root of $ax^3 + bx^2 + cx + d = 0$ is $-\frac{d}{a}$.

Note : Roots of $x^2 + x + 1 = 0$ are complex

Do yourself - 12 :

- Let α, β be two of the roots of the equation $x^3 - px^2 + qx - r = 0$. If $\alpha + \beta = 0$, then show that $pq = r$
- If two roots of $x^3 + 3x^2 - 9x + c = 0$ are equal, then find the value of c .
- If α, β, γ be the roots of $ax^3 + bx^2 + cx + d = 0$, then find the value of

(a) $\sum \alpha^2$

(b) $\sum \frac{1}{\alpha}$

(c) $\sum \alpha^2(\beta + \gamma)$